

MULTIDISCIPLINARY PROJECT
FINAL REPORT

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**ALTA SCUOLA POLITECNICA
MULTIDISCIPLINARY PROJECT
FINAL REPORT**

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1 Executive Summary

Rising temperatures associated with climate change are accelerating glacier retreat throughout the Alpine region, exposing unstable sediment deposits that can be rapidly mobilized during intense precipitation events and as a consequence generate destructive debris flows. The Belvedere Glacier and its surrounding basin represents a significant example of this phenomenon, where sediment instability poses a potential hazard to the downstream area and in particular to the municipality of Macugnaga. Effective risk management requires continuous observations of sediment dynamics, yet conventional monitoring techniques are often expensive, labor-intensive, and difficult to deploy in remote mountain environments. As a result, such limitations highlight the need for an autonomous and reliable monitoring approach capable of providing objective data over extended periods. The SENTINEL project was conceived to address this need by enabling the continuous observation of glacier environments and providing information that can contribute to hazard assessment, scientific research, and informed decision-making.

The first objective is the development of an autonomous monitoring station intended for remote environments. The platform is designed to generate consistent environmental data, providing long-term optical measurements without the requirement of frequent on-site interventions. Another objective is to transform these observations into quantitative information about suspended sediment transport through dedicated image-analysis models which enable the assessment of variations in time of water turbidity. The resulting data are intended to constitute the base to support scientific investigations, improve the understanding of sediment transport processes and provide valuable information for practical population management under critical conditions.

To achieve the project objectives, the monitoring system had to satisfy several technical requirements. The solution must allow autonomous operation through a non-invasive optical monitoring approach, while ensuring long-term reliability under harsh environmental conditions. The supporting structure was required to withstand the combined effects of gravitational and wind loads while remaining maintainable and easily transportable to the installation site without the need for specialized equipment. At the same time it also needs to ensure that the optical system has an adequate image resolution and an appropriate field of view to provide high-quality input data for the image-processing pipeline. These requirements constituted the basis for the subsequent mechanical design and structural validation of the monitoring platform.

To satisfy the identified requirements the developed solution combines an autonomous imaging platform with a dedicated mechanical support designed for long-term deployment in harsh environments. The Tikee imaging system was selected for its autonomous operation and compatibility with the optical requirements. This required a custom mounting system to be designed to ensure the correct camera positioning, maintaining the required field of view and image resolution for the image-processing workflow. Such a structure was developed using commercially available components whenever possible to simplify assembly and maintenance while also balancing structural performance, weight, and cost. In terms of structural integrity the proposed solution was verified through multiple computer simulations (finite element analyses) under both gravitational and combined gravity and wind loading adopting conservative assumptions regarding wind velocities exceeding the maximum historical values recorded at the installation site and surface affected by the wind. Several design iterations were evaluated to progressively optimize the support system, leading to the introduction of a tensioned cable stabilization system that improved structural stability and reduced displacement under both static and dynamic loading conditions. The dynamic behavior of the complete structure was assessed through modal and harmonic response analyses. Comparing the results of these analyses with the environmental sources of mechanical excitation it was

confirmed, under different modeled assumptions, that the predicted structural response remains within the mechanical limits of the selected material under realistic operating conditions. Overall, the developed solution provides a numerically verified monitoring platform that satisfies the identified design requirements. The prototype was manufactured, installed at the Belvedere Glacier site, and successfully commissioned, demonstrating the feasibility of the proposed system for autonomous operation in the target environment.

The images acquired through the monitoring platform are processed through a machine learning based image analysis pipeline developed to estimate quantitatively water turbidity, which is correlated with suspended sediment and sediment instability, directly from optical observations without requiring direct contact with water. At the current stage the model for the turbidity estimation has been validated only under controlled laboratory conditions, demonstrating the feasibility of the proposed approach, however, further field calibration will be required after the system deployment to account for external environmental factors, such as variations in illumination, changing weather conditions, and other seasonal changes affecting the measurement. Under grouped cross-validation on the controlled laboratory dataset, the image-analysis pipeline achieved a mean absolute error of approximately 5.8 percentage points for turbidity estimation and a material-classification accuracy of approximately 93.6%. Only after being validated under real operating conditions, however, the extracted information may be finally integrated with complementary environmental data to support sediment monitoring and future hazard assessment activities.

In conclusion, the SENTINEL project resulted in the design of an integrated monitoring system composed of an autonomous imaging platform, a dedicated mechanical support and a machine learning-based image-analysis pipeline used for water turbidity estimation. The mechanical solution was numerically verified under realistic environmental loading conditions while the image-analysis model demonstrated the feasibility of estimating turbidity from optical images under controlled laboratory conditions. This establishes a solid technical foundation for prototype manufacturing, and subsequent calibration under real operating conditions, representing the first step towards a reliable long-term sediment monitoring system for glacier-fed river environments. Building on these results, SENTINEL provides the basis for further development towards long-term autonomous monitoring in Alpine environments. Future work can focus on long-term operation verification of the monitoring system, allowing the image analysis model to be calibrated and refined using data collected under the varying environmental conditions typical of the installation site and confront it with direct sediment measurements. The integration of turbidity measurements for calibration and other complementary environmental data could further enhance the understanding of sediment dynamics and support future hazard assessment or early warning systems. Such developments would extend the capabilities of the current system from a validated and installed prototype towards an on site tested solution for monitoring glacier environments.

2 Introduction

2.1 Background

It is widely recognized that climate change has a severe impact on nature and Alpine regions are highly vulnerable environments. The geomorphological stability of high-mountain areas is critically endangered by the effects of climatic shifts. Among these, one of the most visible phenomena is the massive shrinkage of mountain glaciers, which inevitably leads to a cascade of environmental issues, as the dramatic increase in slope instabilities [13]. The structural integrity of high-altitude rock walls and lateral moraines (i.e. accumulations of glacial debris carried and deposited by a glacier) can be altered by the combination of the degradation of permafrost (i.e. ground that remains completely frozen for at least two consecutive years) and intense glacial retreat. Therefore, it is possible to notice an increase in frequency of rockfalls, ice avalanches and debris flows.

Within this context, the eastern slopes of Monte Rosa and its debris-covered glacier tongue, the Belvedere Glacier, were selected as a primary test site for understanding complex glacial and paraglacial dynamics. A peculiar behavior was observed in the Belvedere Glacier as unlike most retreating Alpine glaciers, its front was maintained at relatively low elevations, approximately 1800-1900 m above sea level because of a thick debris cover which reduced ablation rates. In the last three decades, the glacier exhibited surge-type movements (i.e. rapid and sudden accelerations of a glacier) and severe morphological transformations. Between 2000 and 2002 a surge event led to the deformation of the glacier surface and the overriding of Little Ice Age lateral moraines [13]. In addition to the above described effects, another consequence of climate change was the evolution of supraglacial and proglacial lakes (e.g., Lago delle Locce and Lago Effimero) which is a threat to downstream infrastructure, in particular to the Macugnaga village, because of a phenomenon called Glacial Lake Outburst Flood (GLOF): a sudden and catastrophic release of a large volume of water from a glacial lake [13][5]. Furthermore, rapid ice down wasting (i.e. the progressive thinning and vertical reduction of a glacier) induced instability along the lateral moraines. This phenomenon triggered continuous moraine sliding, which actively threatened tourist trails and alpine huts [5]. Additionally, in the last few years, extreme weather events took place, triggering massive geomorphological responses. For example, in August 2023 a devastating debris flow mobilized around 180000 cubic meters of material from the Castelfranco slope, creating deep erosional channels into the glacier's western tongue [5][16].

The coupling between climate change, glacial retreat and land alteration must therefore be investigated to understand the mechanism responsible for the increase of such hazardous events. In this context, particular attention was devoted to sediment mobilization and transport within the glacier-fed river system, as variations in water turbidity provide an observable indicator of changes in suspended sediment conditions during intense meteorological and geomorphological events. Monitoring fluvial suspended sediment is essential because such sediment concentration changes in compliance with phenomena happening upstream and, moreover, as an excess of suspended sediment can adversely affect water supply, hydropower generation, fisheries, and flood control [28].

Some of the geomorphological phenomena cited before are characterized by fast transitory development, as for glacial surges and debris flow, the same applies to sediment transport, which is variable throughout the day and the year. Consequently, seasonal or annual surveys are not sufficient to comprehend the evolution of the glacier and, therefore, to anticipate critical phenomena, manage hydrogeological risk and mitigate landslide and floods hazards continuous monitoring is mandatory [13].

Measuring glacier thickness or area directly on the glacier requires on field surveys and it is, therefore, not feasible to continuously monitor these parameters. Monitoring a glacier-fed downstream water course instead of monitoring directly the glacier allows obtaining information on the glacier's conditions in a continuous manner [29].

2.2 Motivation and Stakeholders' needs

Despite the critical need for comprehensive environmental data, the monitoring solutions available so far were heavily limited by logistical, technological, and economic constraints. Traditionally, the monitoring of glacier thickness, area evolution and glacial motion has been done through time-consuming field activities such as point stratigraphy, manual mass balance measurements and radar soundings whose main limitations are the lack of continuous, detailed and accurate spatial information.

More recent solutions rely on remote sensing, utilizing either Uncrewed Aerial Vehicles (UAVs) or Very High Resolution (VHR) satellite imagery. While UAV photogrammetry provides exceptional spatial resolution, it suffers from severe altitude constraints, limited battery performance in cold temperatures, and the absolute necessity of in-situ human deployment [15]. Conversely, VHR satellite constellations (such as Pléiades and Pléiades Neo) eliminate the need for field operators and cover vast, steep accumulation zones [16]. However, satellite tasking is expensive, subject to orbit availability and cloud cover, and frequently suffers from geometric vulnerabilities on a nearly vertical topography of mountain faces like the Monte Rosa [16].

Furthermore, deploying in-situ continuous turbidity is prohibitively expensive and prone to equipment destruction during the extreme flood events [27].

Therefore, current installations face four primary limitations:

- **Accessibility:** it is difficult for human operators to reach high-altitude sites or steep riverbanks safely [16].
- **Power:** the lack of a power grid and the rapid depletion of batteries in sub-zero temperatures severely restrict the operational lifespan of traditional sensors [28].
- **Maintenance:** hardware can be continuously and easily damaged by rockfalls, avalanches, or debris flows, thus requiring frequent maintenance[28].
- **Costs:** commercial satellite and other sensors represent an unsustainable cost for continuous monitoring [16].

2.2.1 Stakeholders' needs

Another reason that justified the design of the system proposed in this work is the strong interest from a wide range of stakeholders, here the main ones and their needs are presented:

- **Civil Protection and Local Inhabitants:** these stakeholders showed interest in understanding hazardous phenomena (such as landslides and avalanches), reducing safety risks for tourists and mitigating the socio-economic impacts of climate shifts on the valley population. These objectives are achievable through the analysis of continuous monitoring data.
- **Regione Piemonte:** this institution required precise knowledge of glacier dynamics to enhance the safety for tourists and locals.

- **Hydroelectric and Agricultural Industries:** these sectors showed interest in the project because of the possibility of forecasting the seasonal water availability. This data plays a crucial role in maximizing downstream energy production and planning irrigation channels.
- **Sensor Manufacturers:** these sectors recognized the extreme alpine environment as a testing ground to develop remote sensing technologies and many companies showed interest in the obtained results.

The motivations for the SENTINEL project are rooted in the current installations' limitations and the strong interest from the stakeholders. The goal is to develop an autonomous, low-cost, and robust monitoring system to remotely track water turbidity and sediment concentration.

2.3 Project Objectives

To overcome the technology limitations and to satisfy stakeholders' needs cited above, the SENTINEL project was established by designing an autonomous optical monitoring system capable of continuously acquiring observations of a glacier-fed river. The scientific objective of the project is converting the acquired optical observations into quantitative information related to water turbidity and, consequently, suspended sediment conditions.

The objectives assigned to the students' team to fulfill external institutions and partners' requirements were structured as measurable engineering goals:

1. **Autonomy:** the primary objective of the monitoring system is that it must not require human intervention during operation.
2. **Long-term operation:** the developed hardware architecture was required to ensure long-term operation. The goal of the system was to be non-reliant on battery replacement, energy-efficient and self-sustaining through solar panels. Furthermore, the system needs to be resistant to extreme weather conditions to avoid continuous and time-consuming maintenance interventions.
3. **Automatic data extraction and transmission:** the system needs to be able to extract and transmit data so that they can be elaborated afterwards to obtain the information needed on changes in water turbidity and suspended sediment conditions.

2.4 Method of Work

For ease of development the human resources have been split into two independent teams:

- **Team Camera processing**
- **Team Infrastructure**

2.4.1 Team Camera processing

The objective of this team was to research and develop an information processing pipeline capable of extracting useful information from an image capture of the target water stream. The method has been validated throughout controlled lab experiments.

The team was responsible to show experimentally that the developed methodology can be used, how its performance is related to the setting and how robust to environment changes it can be.

Members: Ginefra Paolo, Gandini Matteo, Abdelaiz Shahd

2.4.2 Team Infrastructure

The objective of this team was to design the necessary infrastructure to deploy the system on the target stream and grant the technological requirements. The system must:

- autonomously capture at least 4 images per hour
- be continuously powered
- transmit the images to a remote server
- resist the harsh mountain environment
- be able to function under adverse weather conditions
- require low maintenance

The team was also responsible for the organization of a field expedition meant for the deployment of the system shown in Fig 1.

Members: Sampò Jacopo, Mazzola Simone Maria, Dalmazzo Nicolò, Moltrasio Aronne

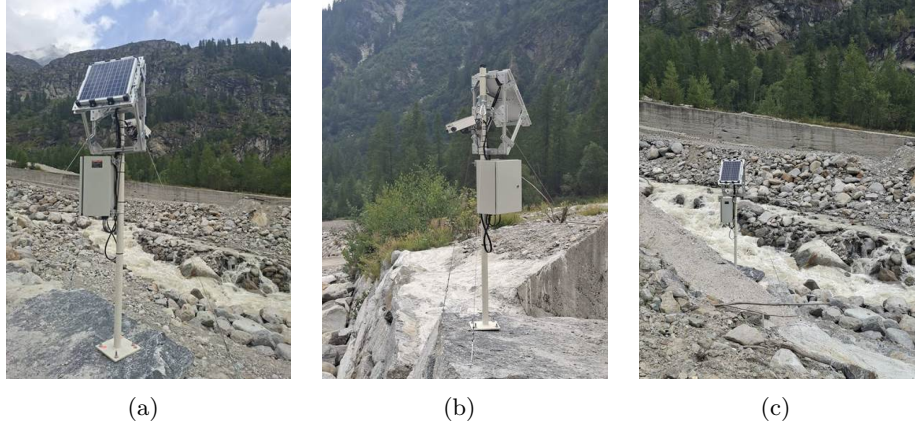


Figure 1: The autonomous SENTINEL monitoring station deployed on-site: (a) Solar panel powering unit. (b) Protective control box. (c) Optical camera pointing towards the alpine stream.

Figure 2 represents the workflow of the two different teams, comprehensive of every task and also how the two different teams interact. The state of the art review on turbidity usage provided useful information also for the system design and the same happened after the data processing experiment, which defined what kind of information was necessary and how to store it. Furthermore, after the information storage was designed by the team Infrastructure, some information processing refinement were made by the Camera processing team. Lastly, the lab experiment was done only after the camera and components were purchased by the other team.

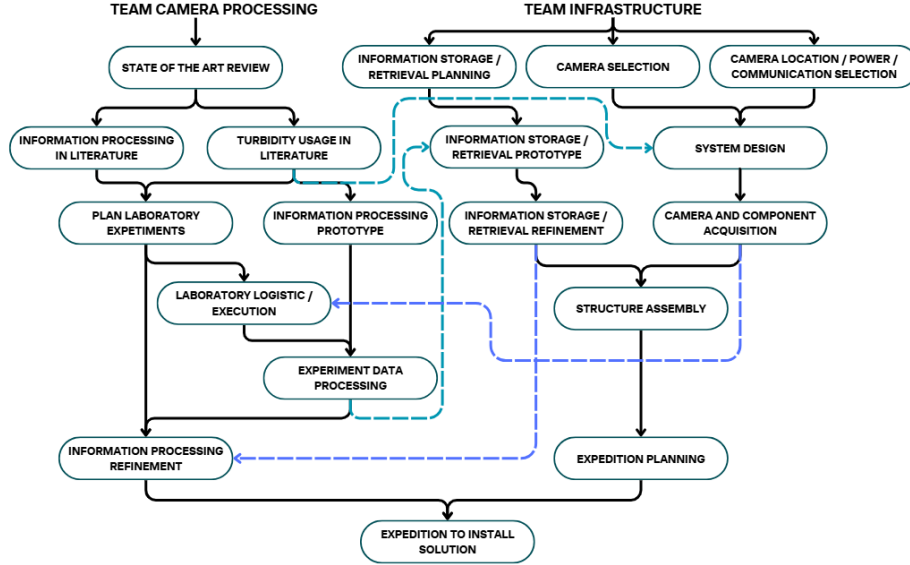


Figure 2: Precedence diagram illustrating the interdisciplinary workflow and task dependencies of the SENTINEL project, divided into Team Camera Processing (left) and Team Infrastructure (right), converging at the field installation expedition. Dotted arrows represent the exchange of information and cooperation among different working groups inside the team.

Regarding the project timeline, the entire workflow was divided into different phases. Throughout the project's realization, monthly meetings were scheduled so that every group member and the academic tutors could be updated with the work. The phases are outlined below:

- **First expedition:** the team gathered information on the geomorphology of the area and its limitations, as well as past accidents' history.
- **Literature review:** the State of the Art and similar projects were reviewed in detail to check which sensors and strategies had shown to be successful and which not. The output of this phase was a common and detailed knowledge of glacier's geomorphology and previously employed solutions for glacier's monitoring purposes.
- **Possible solution exploration:** in this phase all potentially relevant solutions were explored and the non-feasible solution excluded. The initial investigation considered a broader range of alpine monitoring applications, including slope instability and rockfall detection. During the project development, however, the target application was progressively focused on sediment mobilization and water turbidity in the glacier-fed river system, which became the specific environmental quantity addressed by the final monitoring architecture. The output of this preliminary phase was a first concept of the system specific aims.
- **Design of the system:** this part constitutes the core of the work, the hardware concept was designed, the components were purchased and the camera was tested in the lab. In parallel, the

image processing software was designed. During this phase a comprehensive set of verifications was conducted across all subsystems to ensure reliability in harsh alpine conditions. In the energetic domain, energy availability was simulated using PVGIS, while the power balance was verified through sensitivity analyses and dynamic 12-month state-of-charge simulations under worsening operative scenarios. Furthermore, the mechanical structure was validated through finite element method (FEM) simulations while the image-analysis pipeline was verified through a laboratory smoke test in which hundreds of images were acquired across multiple materials, concentrations, and viewpoints. The output of this phase was the hardware and software of the final system, which still needed to be assembled and installed.

- **Deployment of the system on site:** an expedition was planned in order to complete the on site installation and assembly of the system. The output of this phase was the completion of the hardware realization and field installation and the system-level operational verification.
- **Post-processing:** by using a neural network the data acquired with the sensor on site was processed to extract information on the sediment concentration from images. The output of this phase was the completion of the software component of the system and, therefore, the end of the project.

The final part of the work consists of the realization of this final report. Further information about it is given in the following paragraph.

2.5 Report organization

The report is organized as follows: Chapter 3 focuses on the needs of the diverse stakeholders, translating them into quantifiable requirements. Chapter 4 reviews the state of the art in slope and mass movement monitoring, examining time-lapse optical photography, commercial solutions and autonomous off-grid power systems, thereby highlighting the existing technological gaps. Chapter 5 presents the SENTINEL monitoring system, covering: the reasons behind the selection of the camera, the sizing methodology of the electrical system, the mechanical design through structural finite element simulations and dynamic verifications, the image processing based on machine learning for turbidity estimation and the final on-site installation. Finally, Chapter 6 draws the conclusions of the project, summarizing the achieved outcomes, discussing current experimental limitations, and outlining guidelines for future field validation and long-term hazard assessment.

3 Users' Requirements

This chapter translates the needs identified for the monitoring application into system-level requirements for the SENTINEL platform, defining what the system must achieve, the environmental and operational conditions under which it must remain functional while leaving the specific technological implementation to the subsequent design phase.

The monitoring of glacier-fed rivers, in particular for the observation of sediment dynamics and water turbidity in environments such as the Belvedere Glacier, represents a demanding application because observations must be obtained continuously in a remote and environmentally harsh setting. The geomorphological stability of high-mountain areas is critically affected by climatic changes leading to glacial retreat, permafrost degradation, and potentially more frequent rockfalls and debris-flow events. In this context conventional monitoring approaches face significant economic, environmental, and logistical limitations as field surveys are time-consuming and expose operators to hazardous conditions while approaches requiring frequent human intervention are difficult to sustain in remote alpine environments. Remote and non-invasive monitoring is therefore required to provide sufficiently frequent observations while limiting maintenance and physical access to the site and, as such, the requirements defined in this chapter consequently focus on the required monitoring capability, data availability, autonomy, robustness, maintainability, and environmental compatibility of the system.

The methodology used to depict and consolidate these requirements follows a rigorous system engineering approach, combining empirical site assessments with stakeholder analysis. Initial operational boundaries were derived from a comprehensive literature review of historical high-altitude monitoring challenges and prior glaciological studies and were subsequently refined through feedback from the relevant stakeholders, preliminary on-site surveys and topographical analyses at the Belvedere Glacier. The resulting requirements define the functional capabilities and performance constraints that the subsequent design must satisfy.

3.1 Stakeholders

Understanding the stakeholders involved in the SENTINEL project plays an important role, as each of them brings distinct expectations and constraints. Furthermore, their combined needs shape all the subsequent technical requirements.

In Fig 3 it is possible to find some of the stakeholders involved in the process of developing of the SENTINEL project and the relationships they present.

Researchers rely on the system here presented to improve the understanding of sediment transport processes and geomorphological alterations. For these institutions, seasonal or annual surveys are insufficient to characterize short-term variations and event-driven changes, creating a need for observations with adequate temporal continuity and spatial detail. The monitoring platform must therefore provide consistent and sufficiently informative observations over extended periods, while ensuring that the acquired data remain suitable for subsequent scientific analysis. In this complex context, it is important to maintain a hierarchy of relevance of the stakeholders and the ultimate beneficiaries of the monitoring system that are: Civil Protection authorities, Arpa, local inhabitants of the Macugnaga municipality, and regional institutions like Regione Piemonte. All these institutions depend on continuous data availability even during extreme meteorological events to monitor changes in environmental conditions associated with potentially hazardous geomorphological and hydrological events and to support subsequent assessment and decision-making.

Value chain positioning-diagram

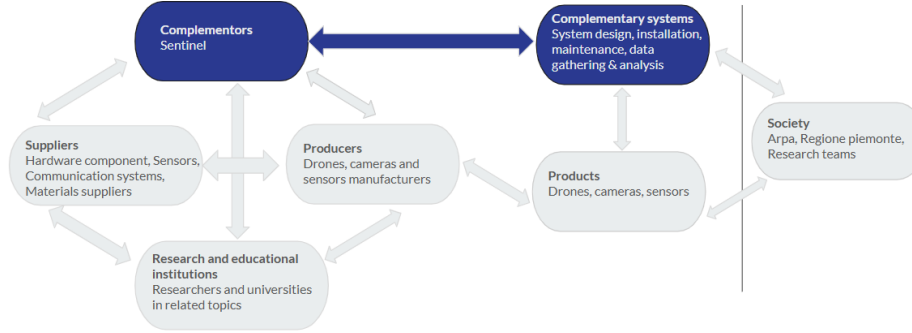


Figure 3: Value chain positioning-diagram of our stakeholders with their relationships.

Additionally, the project attracts interest from hydroelectric and agricultural industries who may benefit from improved understanding of sediment and water conditions in glacier-fed catchments, particularly where changes in sediment transport can affect downstream infrastructure and water-management activities. Sensor manufacturers also view this extreme alpine environment as a valuable testing ground to develop robust remote sensing technologies, creating new industrial opportunities ultimately being interested in the results such a system will be able to yields.

From an institutional perspective, Politecnico di Milano requires the successful demonstration of a scientifically rigorous engineering approach. The project is expected to provide a technically justified solution that can serve as a basis for future monitoring campaigns and further development. Finally, the installation crew and maintenance operators responsible for deploying and maintaining the physical infrastructure require a system that prioritizes safety and accessibility. Deploying hardware in high-altitude environments exposes operators to significant risks, mandating a system that is easily transportable without specialized heavy equipment and highly modular to allow for rapid, localized troubleshooting without complete system disassembly.

3.2 Functional Requirements

The functional requirements define the core operational capabilities the monitoring platform must actively execute to satisfy the overarching project objectives. Primarily, the system shall provide periodical observations of the monitored riverbed according to a reliable and consistent acquisition schedule while also capturing images with a sufficiently wide field of view to cover all the necessary river section and maintain a sufficient image resolution to preserve the necessary ground resolution to allow changes in the monitored environment to be identified over time. To ensure temporal continuity for geological mapping, the system must provide sufficiently frequent image acquisition to capture relevant changes in the monitored environment prompting the acquisition frequency to be selected based on this monitoring requirement and considering the trade-off between temporal resolution, energy availability, communication bandwidth, and storage capacity. The resulting operating schedule must take into account that the selected system will have to operate during daylight hours to obtain observations of adequate quality, adapting to the natural

illumination of the site.

Given the logistical difficulty and safety implications of retrieving data manually from the deployment site the monitoring platform must provide remote access to the acquired observations without requiring routine physical intervention. The communication system has to provide sufficient capacity and reliability to transfer the generated monitoring data and support continuous remote data availability while also, in the future, allowing the monitoring capability to be extended towards more time-critical applications if required. To achieve this, the station must operate entirely autonomously. It must automatically adapt its schedule to local sunrise and sunset conditions, exploiting natural illumination while completely avoiding unnecessary acquisitions and energy consumption during nighttime hours. The platform must manage its operating state so as to minimize unnecessary energy consumption while also preserving the required monitoring functionality hence the energy consumption between acquisition events has to be minimized without compromising the required availability and reliability of the system. Reliability is critical, the system must seamlessly recover from transient errors and include appropriate mechanisms to preserve data continuity. Finally, the physical architecture must allow safe and practical installation at the designated deployment site while minimizing the need for specialized equipment and extensive on-site intervention.

3.3 Non-functional Requirements

Non-functional requirements establish the quality attributes and operational boundaries of the monitoring platform. Power independence is a fundamental requirement for the monitoring platform as the system must operate autonomously over extended periods without relying on permanent external electrical infrastructure or frequent manual intervention. The energy subsystem has to provide sufficient energy availability to sustain the required monitoring and communication functions under the expected environmental conditions, including periods of reduced renewable-energy availability. The energy architecture must therefore be designed with sufficient generation and storage capability to maintain the required level of service during worsening conditions.

Economically, the platform must represent a low-cost alternative to traditional monitoring techniques while maintaining the required monitoring performance, favoring solutions that are economically sustainable over long-term deployment and maintenance. Maintainability has to be supported through a modular architecture that allows maintenance activities and component replacement to be performed with minimal disassembly and limited on-site intervention. Safety and robustness are paramount. Electrically, the installation must include protection on both the energy-generating input and storage system's supply lines to guard against accidental short circuits and wiring faults. Mechanically, structural deformation and vibration shall remain sufficiently limited to preserve the stability and repeatability of the observation geometry under expected environmental loads. The structural system must therefore withstand the expected environmental loads with an appropriate safety margin and without compromising the required operational performance. Moreover, the system must also provide adequate protection against precipitation, moisture, snow, corrosion, and other environmental exposure expected throughout the year at the deployment site. Ultimately, a modular organization of the various subsystems needs to be developed, ensuring scalability and allowing the system to serve as a replicable blueprint for different glacier-fed river environments.

3.4 Environmental Constraints

The alpine deployment site imposes severe environmental constraints that act as primary drivers for the entire engineering process. The steep topography and geomorphological instability of the mountain

environment, together with the risk associated with floods, debris flows, and sediment transport, make permanent submerged instrumentation particularly difficult to maintain and access. These conditions impose the need for monitoring approaches that minimize physical intervention in the river and reduce the frequency of direct site access. Wind loading represents a significant environmental constraint for the supporting structure. The mechanical system has to be capable of withstanding the expected extreme wind conditions at the deployment site with an appropriate safety margin. Furthermore, structural deformation and dynamic response also have to remain sufficiently limited to preserve the stability of the observation geometry and avoid adverse effects on image acquisition. As such, potential resonance phenomena associated with wind-induced excitation must therefore be considered during the structural design and verification.

Snow accumulation have to be considered in the design of the energy-generation subsystem, with the system configuration ensuring adequate winter performance and minimizing the risk of prolonged loss of energy production caused by the stagnation of snow and ice. The energy-storage subsystem has to remain within its safe operating conditions under the expected temperature range of the deployment site, particularly, the system needs to prevent operating conditions that could cause degradation or unsafe charging of the energy-storage technology during prolonged sub-zero temperatures. Thermal management must therefore be considered in the design of the energy subsystem to separate the system from the outside meteorological conditions. Solar-resource variability, including the effects of the surrounding alpine topography and seasonal shading, must be appropriately addressed when defining the energy requirements of the autonomous monitoring system. The energy subsystem needs to maintain the required monitoring capability during periods of reduced solar availability.

Taken together, these requirements define the performance, autonomy, reliability, maintainability, and environmental robustness that the monitoring platform must achieve. The following chapters use these requirements as the basis for evaluating available technologies and deriving the final system architecture.

Requirement	Type	Derived specification / target	Design outcome
Continuous environmental monitoring	Functional	Sufficient temporal sampling	15-min acquisition
River-section observation	Functional	Sufficient spatial coverage	~8 m coverage
Informative optical observations	Functional	Adequate ground resolution	~1 cm
Remote data availability	Functional	Automatic wireless transmission	LTE
Long-term autonomy	Non-functional	Energy-positive operation over expected conditions	PV + battery
Energy availability	Non-functional	Sufficient generation and storage	20 W + 144 Wh external subsystem

Requirement	Type	Derived specification / target	Design outcome
Structural stability	Non-functional	Limited deformation and dynamic response	Guy-wire reinforced structure
Environmental robustness	Environmental	Operation under expected alpine conditions	Enclosure, insulation, protected components
Maintainability	Non-functional	Minimal intervention and modular replacement	Modular electrical/mechanical architecture

Table 1: System requirements and design outcomes.

4 State of the Art

Mountain environments exposed to rockfall and slope instability have historically been difficult to monitor because of their remoteness, the absence of grid power, and the harsh climatic conditions typical of alpine terrain. Over the past two decades, advances in remote sensing, low-power electronics, and renewable energy have made continuous, autonomous monitoring of such environments increasingly feasible. This chapter reviewed the main techniques used to detect and characterize rockfall activity, examined time-lapse photography as a complementary monitoring method, surveyed existing commercial solutions, and discussed the autonomous power systems required to operate monitoring stations in off-grid conditions.

The review initially considered monitoring technologies developed for a broader range of alpine geomorphological hazards, including slope instability and rockfall, as these represent closely related requirements in terms of remote operation, autonomy and environmental robustness. As it is discussed in section 4.5, the final application considered in this project is, however, focused on optical monitoring of turbidity and suspended sediment dynamics in a glacier-fed river.

4.1 Rockfall Monitoring

Rockfall monitoring techniques can be grouped into geodetic and remote sensing methods, contact-based instrumentation, and, more recently, distributed wireless sensor networks (WSNs). Each category offered a different trade-off between spatial coverage, temporal resolution, accuracy, and cost.

Terrestrial Laser Scanning (TLS) has been one of the most widely adopted remote sensing techniques for rock slope characterisation. Jaboyedoff et al. reviewed the use of light detection and ranging (LIDAR) in landslide studies and classified its applications into detection and characterisation of mass movements, hazard and susceptibility assessment, modelling, and monitoring, noting that high-resolution digital elevation models derived from LIDAR opened new domains of investigation that were, at the time, still under development [17]. Abellán et al. later reviewed the specific application of TLS to rock slope instabilities, covering deformation detection at sub-centimetric to millimetric scales and change-detection workflows between successive scans [1]. TLS surveys typically require repeat campaigns every few weeks or months, which limits their ability to capture events that unfold on shorter timescales, as illustrated in a baseline monitoring program along a transportation corridor in the Fraser River valley, where scans were repeated every two to four months to track precursory signs of failure [20].

Ground-based radar interferometry (GB-InSAR) complements TLS by providing continuous displacement measurements with millimetric accuracy over an entire slope, without requiring physical contact with the rock mass. A study on the Hongshiyuan post-earthquake rock slope combined TLS and GB-InSAR to reconstruct the full failure process of a rockfall collapse, using the laser-derived three-dimensional model to identify discontinuity sets while the radar tracked the pre-failure displacement evolution [22]. A more recent study applied the same combination, together with ambient vibration analysis and rockfall runout modelling, at the Temple of Hatshepsut in Egypt, arguing that no single observation method could account for the whole range of processes that prepare and trigger rockfall, and that complementary methods were still needed even after decades of progress in the field [18].

Wireless sensor networks represent a more recent and lower-cost alternative, particularly suited to sites where wiring is impractical or where budgets do not allow for radar or laser-based systems. A comprehensive review organized over one hundred journal articles on WSN-based rockfall and landslide monitoring and identified low-power wide area network (LPWAN) technologies, in particular LoRa, as

the most frequent choice of radio communication because of their favorable trade-off between range and energy consumption [33]. The same review reported that early prototypes already achieved useful results with inexpensive microcontrollers, commercial sensors, and low-power transmitters, and that the field had since moved toward more efficient implementations, including energy harvesting to extend the operational life of nodes deployed in inaccessible areas [33]. Building on this trend, a smart autonomous LoRa-based system was deployed on Pantelleria Island, integrating a triaxial MEMS accelerometer to estimate slope tilt and a solar-powered gateway assembly to provide connectivity independent of fixed infrastructure [32]. A subsequent design used ESP32 microcontrollers together with SX1276 LoRa transceivers, MEMS accelerometers, and GPS modules in a low-cost prototype aimed specifically at rockfall and landslide monitoring in mountainous terrain [31], while a further multi-nodal architecture proposed a custom communication framework to improve reliability across a network of distributed nodes [11].

Acoustic emission (AE) monitoring and infrared thermography have also been explored as complementary techniques. Acoustic sensors can detect the micro-fracturing that precedes slope failure, offering an early indication of instability, although installing contact sensors on steep, unstable rock faces can itself be hazardous and costly [8]. Infrared thermography, by contrast, measures surface radiant temperature remotely and can reveal moisture content and thermally driven deterioration relevant to rockfall susceptibility, without requiring any contact with the monitored surface [33].

Table 2 summarizes the main advantages and limitations of the techniques discussed above.

Technique	Spatial coverage	Temporal resolution	Typical accuracy	Main advantages	Main limitations
Terrestrial Laser Scanning (TLS)	Whole slope, high point density	Campaign-based (weeks to months)	Millimetric to sub-centimetric	Detailed 3D geometry, non-contact	Discontinuous in time, requires line of sight, costly equipment
GB-InSAR / GB-SAR	Whole slope	Continuous, near real-time	Millimetric	Continuous displacement tracking, non-contact	High equipment cost, sensitive to atmospheric noise
Acoustic emission (AE)	Local, sensor-dependent	Continuous	Event detection, not geometric	Detects precursory micro-fracturing	Contact-based, hazardous installation, limited spatial coverage
Infrared thermography	Whole slope	Campaign-based or continuous	Qualitative to semi-quantitative	Non-contact, reveals moisture and thermal anomalies	Affected by weather, limited depth penetration

Technique	Spatial coverage	Temporal resolution	Typical accuracy	Main advantages	Main limitations
Wireless Sensor Networks (WSN/LoRa)	Distributed point measurements	Continuous, near real-time	Millimetric tilt/displacement (node-dependent)	Low cost, low power, no wiring required, scalable	Limited spatial density, battery/solar dependency, network reliability in harsh terrain

Table 2: Comparison of rockfall monitoring techniques.

4.2 Time-lapse Monitoring

Time-lapse photography, defined here as the automated, repeated acquisition of images from a fixed viewpoint, has been used in geosciences for well over two decades, originally in glaciology and later extended to a wider range of geomorphological processes. Its main advantage lies in the low cost of the equipment relative to laser scanning or radar interferometry, combined with the possibility of continuous, unattended operation over long periods.

A review of terrestrial monoscopic time-lapse cameras applied to glacier flow surveying reported early applications dating back to the early 2000s and noted a high benefit-to-cost ratio compared with alternative survey methods, while also cataloguing case studies across the European Alps, Scandinavia, Iceland, Patagonia, Greenland, and Svalbard [9]. The same review observed that terrestrial digital cameras had, in the last two decades, been extended beyond glaciology to monitor rock glaciers, landslides, rivers, volcanoes, and snow cover, indicating that the technique had become a general-purpose tool for geomorphological monitoring rather than a glaciology-specific method [9].

Applications to mass movements more directly relevant to rockfall monitoring include a study of the Ritigraben rock glacier in the western Swiss Alps, which combined georeferenced time-lapse photography with digital image correlation to track surface displacement, reporting that monoscopic time-lapse photography had proved to be a reliable monitoring method for an alpine mass movement with temporally consistent movement directions and limited changes in overall geometry [21]. A separate study on the Tapado Glacier in the Dry Andes of Chile used daily time-lapse image sequences, orthorectified against a digital terrain model, to document albedo variation and to track supraglacial ponds and ice cliffs, describing the approach as a cost-effective method for continuous observation in a remote high-altitude setting [39]. More recently, an AI-based tracking method applied to high-frequency monoscopic time-lapse imagery was used to monitor fast-moving alpine landforms, including active rock glaciers and landslides, and was described as one of the first applications of artificial intelligence methods to this type of image data in geomorphology [14].

The main limitations of time-lapse monitoring are related to the two-dimensional nature of a single camera viewpoint, the sensitivity of image correlation algorithms to changing illumination, shadows, and reduced visibility, and the need for georeferencing procedures, such as monoplotted onto a digital terrain model, to translate pixel displacements into physical measurements [26]. Feature tracking over

long timescales can also require periodic updating of the correlation template as the monitored surface evolves, along with manual verification of the tracked features [19]. Despite these constraints, the low power and bandwidth requirements of time-lapse cameras make them particularly attractive for autonomous, solar-powered stations, where the volume of data to transmit is far smaller than that produced by continuous radar or laser scanning.

4.3 Existing Commercial Solutions

Commercial monitoring systems for mountain hazards can be broadly divided into three categories: image-based systems and integrated radar-based early-warning platforms. All three categories increasingly rely on wireless communication and autonomous power supply to operate in remote terrain.

Among image-based systems, robotic gigapixel camera platforms, such as the Gigapan system paired with a DSLR camera, have been deployed at experimental sites to obtain high-resolution, multi-temporal photo-monitoring datasets. At the Poggio Baldi Landslide Natural Laboratory in Central Italy, a permanent site jointly established by Sapienza University of Rome and the spin-off company NHAZCA, a gigapixel camera was combined with a ground-based ArcSAR radar and an acoustic noise monitor during an intensive three-day campaign, during which the image-based system alone identified 84 individual rockfall events [35]. The laboratory became a fully energy-independent, remotely controlled permanent facility in October 2021, illustrating how commercial and research-grade image-based sensors are increasingly packaged together with autonomous power supplies for continuous, unattended operation [35].

Radar-based early-warning platforms represent a further category of commercial solution. The Geoprevent system, developed by the Swiss company Geobruagg, has been deployed at multiple alpine sites, including a road near Brienz in the Swiss Alps, where a radar mounted roughly one kilometre from the slope detects falling boulders in real time and can trigger automatic road closures [12]. According to the company, monitoring frequency can be adjusted from monthly or weekly intervals under stable conditions to continuous, twenty-four-hour analysis once motion is detected, with built-in redundancy in both power supply and communication links [12].

Geotechnical sensor networks form the third category. Worldsensing, a company specialising in wireless geotechnical monitoring, has deployed wireless tiltmeters connected through LoRa gateways at multiple sites, including landslide-prone residential areas in Malaysia, where four tiltmeters were reported to take readings up to 42 times per day and to operate for months without on-site maintenance [41]. A separate deployment in a high-alpine environment for rockfall hazard mitigation was described by the operating company as functioning reliably despite the challenging conditions, while removing the need for cabling and reducing the exposure of maintenance personnel to rockfall danger zones [41].

Table 3 summarizes representative characteristics of these three categories of commercial solution.

Category	Representative example	Communication	Power supply	Typical use case
Image-based (gigapixel / time-lapse camera)	Gigapan robotic DSLR system	Local storage, periodic transmission	Solar with battery buffer	Photo-monitoring and change detection, low bandwidth

Category	Representative example	Communication	Power supply	Typical use case
Radar-based early warning	Geoprevent (Geobruigg)	Continuous radio/cellular link	Redundant power and communication	Real-time boulder detection, automatic warning/closure
Geotechnical WSN	Worldsensing wireless tiltmeters	LoRa to gateway	Long-life battery, solar-assisted	Slope deformation and building integrity monitoring

Table 3: Comparison of commercial monitoring solution .

4.4 Autonomous Power Systems

Monitoring stations located in mountain environments without grid access depend on photovoltaic (PV) systems combined with rechargeable batteries to operate continuously throughout the year, including during periods of reduced solar irradiance in winter.

Solar panels. Three main PV cell technologies are commercially available: monocrystalline, polycrystalline, and thin-film. Monocrystalline cells, produced from a single continuous silicon crystal, generally reach the highest conversion efficiency, commonly cited in the range of 20 to 24 percent, and therefore deliver more power per unit area, which is a relevant factor where panel size and weight must be minimized for transport to remote sites [40]. Polycrystalline panels, made from multiple silicon crystals, are typically less expensive to manufacture but reach lower efficiencies, generally between 15 and 18 percent; a comparative field study found that polycrystalline modules could nonetheless perform better than monocrystalline modules during periods of high ambient temperature [4]. Thin-film technologies offer the lowest efficiency of the three but tend to be lighter, more flexible, and less sensitive to partial shading, which can make them suitable for irregular mounting surfaces despite requiring a larger surface area to achieve the same power output [3].

Battery technologies. Lead-acid batteries remain in use in some off-grid installations for their low upfront cost, but they lose a substantial share of their usable capacity at low temperature and have a comparatively short cycle life. Lithium-based chemistries, particularly lithium iron phosphate (LiFePO₄), have become the preferred choice for remote and cold-climate installations because of their thermal stability, longer cycle life, and better retention of capacity at low temperature compared with other lithium chemistries such as nickel manganese cobalt oxide (NMC) (industrial power system documentation, 2026). Charging any lithium-based battery below freezing point carries a risk of lithium plating on the anode, which can permanently reduce capacity, so systems deployed in alpine environments generally require either a charge cut-off below 0 °C or a battery management system capable of pausing or limiting charge current under cold conditions [43].

Maximum Power Point Tracking (MPPT). Because the power output of a PV panel varies non-linearly with irradiance, temperature, and load, charge controllers use tracking algorithms to keep the panel operating close to its maximum power point. The two most widely implemented algorithms are

Perturb and Observe (P&O) and Incremental Conductance (INC). Rajani and Pandya compared the two methods under varying atmospheric conditions and found that the incremental conductance algorithm performed better than perturb and observe when irradiance changed rapidly [34]. A later comparison under low irradiance and low temperature conditions found that INC outperformed P&O when irradiance fell below 200 W/m^2 , while P&O reached a higher efficiency, above 99 percent, under low-temperature but stable irradiance conditions, suggesting that the optimal algorithm choice depends on the specific environmental conditions expected at the installation site [2]. Compared with simpler PWM charge controllers, MPPT controllers extract more usable energy from the same panel area, particularly under partial shading, low light, or non-optimal panel orientation, all of which are common in mountain terrain where panels cannot always be installed with an unobstructed, ideally oriented view of the sky [38].

Low-power devices and communication. The choice of communication protocol strongly affects the energy budget of a remote station. LPWAN technologies such as LoRaWAN and Narrowband IoT (NB-IoT) are the two most commonly compared options for long-range, low-power telemetry. A quantitative comparison of the two technologies found that NB-IoT devices generally showed higher energy consumption than LoRaWAN devices, an effect that became more pronounced as the number of end devices in the network increased [25]. A broader review of LPWAN energy consumption across LoRaWAN, DASH7, Sigfox, and NB-IoT similarly found that LoRaWAN and DASH7 achieved better energy performance, largely because of the amount of time end devices could spend in a low-power sleep state [36]. These findings are consistent with the LoRa-based architectures discussed in section 4.1, and they support the wider adoption of LoRa/LoRaWAN in remote environmental and geohazard monitoring, where both range and energy efficiency are critical constraints [32][36].

Table 4 compares the main technology choices relevant to the design of an autonomous power system.

Category	Option	Efficiency / performance	Cost	Suitability for cold, remote sites
PV cell	Monocrystalline	20–24%	Higher	High power density, good for space-limited mounts
PV cell	Polycrystalline	15–18%	Lower	Adequate, slightly better in high heat
PV cell	Thin-film	13–18%	Variable	Lightweight, shade-tolerant, larger area needed
Battery	Lead-acid	Low cycle life	Lowest	Poor cold performance
Battery	Li-ion (NMC)	High energy density	Medium	Limited cold-charge tolerance
Battery	LiFePO4	Long cycle life	Medium–high	Best thermal stability, requires charge cut-off near 0°C

Category	Option	Efficiency / performance	Cost	Suitability for cold, remote sites
MPPT algorithm	Perturb & Observe	Simple, fast under stable conditions	Low implementation cost	Suitable for slowly varying irradiance
MPPT algorithm	Incremental Conductance	More accurate under rapidly varying irradiance	Higher implementation cost	Better for cloudy, variable mountain weather
Communication	LoRaWAN	Lower energy per bit, long range	Low (unlicensed spectrum)	Well suited to remote, off-grid deployment
Communication	NB-IoT	Higher throughput, higher energy use	Licensing/subscription costs	Better urban penetration, less ideal for battery-only nodes

Table 4: Comparison of solar, battery, and communication technologies for autonomous monitoring .

4.5 Discussion

None of the techniques reviewed in this chapter combined every property the intended monitoring station needs at once. TLS and GB-InSAR delivered millimetric precision, but their cost, power draw, and acquisition pattern, periodic campaigns for TLS, continuous high-power operation for GB-InSAR, restricted them to sites with substantial infrastructure and budget. Radar-based platforms such as Geoprevent achieved real-time detection with built-in redundancy in power and communication, and geotechnical networks such as those deployed by Worldsensing achieved dense, low-maintenance coverage of slope deformation. Both, however, were built around a single sensing modality, radar or tilt, rather than continuous optical observation of a site. Wireless sensor networks and time-lapse camera systems, particularly the LoRa-based, solar-powered architectures discussed in Sections 4.1 and 4.2, came closest to the required balance of low cost and low power, however, the literature offered few examples of a station that paired continuous image acquisition with the autonomous power budget and remote data transmission needed for a glacier-fed Alpine catchment, where grid access, maintenance visits, and even basic road access are unavailable for much of the year.

Existing technologies provide, individually, high accuracy, continuous monitoring, or low-power operation, but no single approach reviewed here addressed the requirements of continuous optical observation, low energy consumption, remote data transmission, limited maintenance, and deployment in a remote, glacier-fed Alpine environment all at once. This gap motivates the development of an integrated autonomous imaging platform for monitoring water turbidity and suspended sediment dynamics, bringing together the solar and low-power communication architectures reviewed in Section 4.4 with the continuous optical observation approach reviewed in Section 4.2.

The identified limitations motivate the proposed solution presented in Chapter 5.

5 The Solution

5.1 System Overview

5.1.1 Overall Solution

Rockfall monitoring in remote alpine environments requires more than the deployment of a simple imaging device. The effectiveness of a monitoring campaign is influenced by the ability to constantly acquire high-quality observations, guarantee autonomous operation under harsh environmental conditions in the long term, transmit the collected data and transform such information into meaningful outputs useful for geological analysis. This is the reason why the solution proposed within this project was conceived as a complete monitoring system rather than a standalone hardware installation.

The proposed solution integrates multiple subsystems that operate together to create an uninterrupted monitoring workflow. Firstly, at the physical level, an autonomous monitoring station performs image acquisition, energy generation and storage and structural support. The images collected are then transmitted through a communication network to a cloud-based infrastructure for storage, management and analysis. Finally, thanks to dedicated image processing algorithms based on Machine Learning techniques the acquired data can be processed to extract the desired information (Fig. 4).

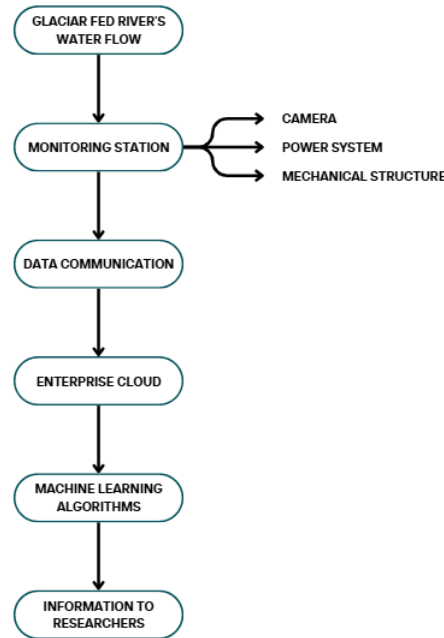


Figure 4: Overall monitoring workflow showing the physical and digital components involved in the proposed monitoring system.

By combining the two aspects of autonomous acquisition system and automated analysis protocols the proposed solution extends beyond the acquisition of images and consists in a monitoring pipeline aiming at reducing maintenance needs while improving the efficiency of data collection, allowing researchers to focus

on the analysis of geological phenomena rather than on the operation and maintenance of the infrastructure used for the monitoring act itself.

Although each subsystem aims at performing a specific function, the overall performance of the monitoring solution depends on their integration and for this reason the system architecture is first presented as a whole before analysing each subsystem individually.

5.1.2 Overall Architecture

Having introduced the proposed monitoring solution, from a systems engineering perspective the solution is organized into two tightly interconnected architectural layers, namely the physical monitoring system and the digital processing infrastructure whose integration, as previously discussed, is essential to achieve continuous autonomous monitoring and transform raw observations into useful information for geological investigations.

Rather than being organized around individual hardware components, the proposed architecture was developed through a functional decomposition of the overall monitoring objective. Such an approach separates the monitoring task into independent functional domains, allowing each engineering discipline to address a specific aspect of the system while maintaining a common operational objective. Each function is assigned to a dedicated subsystem, however, it remains closely interconnected with the others in order to ensure that the overall system satisfies the project requirements.

This functional organization also highlights the multidisciplinary nature of the proposed solution. Due to this aspect, each design decision associated with one subsystem directly affects the performance of the others, making system integration a central aspect of the engineering process and consequently leading to the necessity of considering these interactions when developing the final architecture. Electrical, energetic, structural and computational elements need to be optimized as a whole deployable system rather than independently and, as a consequence, the final design results from a series of engineering trade-offs in which modifications affecting one subsystem inevitably influence the performance of the others.

The resulting architecture establishes the framework upon which the detailed engineering design is developed. Hence, before examining the individual subsystems, it is useful to understand how information and energy flow across the complete monitoring solution during normal operation.

5.1.3 Operational Workflow

The operation of the proposed monitoring solution is based on the continuous interaction between two complementary processes: the management of energy required to sustain autonomous operation and the generation of information that supports geological investigations. Although these processes perform different functions, they evolve simultaneously throughout the monitoring cycle and together define the overall behaviour of the system.

The operational cycle begins with the availability of solar energy, which is converted into electrical power, conditioned and stored in order to ensure the continuous operation of the monitoring station under varying environmental conditions throughout the daily and seasonal cycles. The generated energy is used to supply the imaging subsystem, enabling periodic image acquisition following an a priori defined monitoring schedule while maintaining the autonomy required for long-term deployment. The physical infrastructure therefore provides the conditions necessary for uninterrupted data acquisition independently of external power infrastructures. In parallel the information flow starts with the data acquisition through

the imaging subsystem. Each image is transmitted through the communication infrastructure to a remote cloud environment where it is securely stored and made available for automated processing where Machine Learning algorithms analyse the collected observations to extract quantitative information relevant to the scope of the study. The processed results are subsequently made available to researchers, allowing continuous monitoring activities without requiring permanent on-site presence. At the current stage, the complete field end-to-end chain has not yet been experimentally validated as LTE transmission remains laboratory-verified while the image-analysis model has been validated using controlled laboratory images. The integration of these elements therefore represents the intended operational architecture, whose long-term field performance remains subject to subsequent validation.

The interaction between the energy and information flows further shows the complexity in the interconnection characterizing the proposed solution and for this reason understanding this operational workflow provides the basis for analyzing the engineering solutions adopted for each subsystem in the following sections.

5.1.4 Design Philosophy

Throughout the whole design process of the monitoring solution a set of design principles derived from the operational requirements and environmental constraints has been considered the focal point ultimately guiding the process and providing a common framework for evaluating alternative solutions, supporting the engineering decisions discussed in the following sections. Rather than selecting technologies independently, the development process from the beginning started with the aim of identifying an overall system capable of satisfying multiple objectives simultaneously that could later be subdivided into different compartments.

The primary design objective was to achieve long-term autonomous operation while maintaining a high level of reliability under harsh alpine conditions. Since regular maintenance activities are often difficult, time-consuming and costly in remote environments, the monitoring station was designed to minimize external intervention while also ensuring continuous operation and in order to achieve this goal both the physical and digital architectures were developed to maximize operational continuity and system resilience. A second guiding principle was the integration of complementary engineering disciplines into a coherent and modular system. Structural, electrical, energy and computational aspects were considered from the beginning as interdependent elements of a single monitoring solution rather than as isolated design problems facilitating system optimization but also improving maintainability, allowing future upgrades and subsystem replacements without requiring substantial modifications to the overall architecture. Such modular organization of the system also facilitates scalability, allowing the proposed solution to be adapted or replicated in different monitoring scenarios with limited engineering modifications. These design principles constitute the foundation of the developed engineering solutions that is presented throughout the remainder of this chapter. The following sections, indeed, illustrate how each subsystem was developed to satisfy the identified requirements while contributing to the performance of the monitoring solution as a whole.

Having established the design philosophy underlying the proposed monitoring solution, the remainder of this chapter progressively examines the engineering process that led to its development. The discussion begins with the definition of the system specifications derived from the identified requirements, followed by the design and analysis of each major subsystem, including the imaging, energy, mechanical and electrical architectures. The chapter then presents the integration of these elements into the final monitoring station together with the adopted image processing pipeline. Finally, numerical analyses, experimental validation

activities and overall system evaluation are presented to demonstrate the feasibility and effectiveness of the proposed solution.

5.2 System specifications

5.2.1 Purpose of system specification

The requirements presented in the previous chapter define the needs identified by the stakeholders and the operational context in which the monitoring solution is expected to operate. However, the identified needs alone, due to their high level definition, do not directly provide sufficient guidance for the engineering design process. Before selecting technologies or developing individual subsystems they need to be translated into a coherent set of system specifications able to describe the capabilities the proposed monitoring solution is required to achieve. The requirements identified for the SENTINEL monitoring station include both direct system-level requirements and derived design parameters. System-level requirements define the performance and operational objectives that the proposed solution must achieve, whereas derived parameters are subsequently obtained from these higher-level requirements and from the physical characteristics of the deployment site, and are used to support the quantitative sizing and configuration of the individual subsystems. The complete set of requirements and derived design parameters used throughout the subsequent design process is summarized in Table 5.

The system specifications presented in this section define the expected capabilities and operational behavior required to satisfy the project objectives and therefore encompass the complete monitoring pipeline including both physical and digital aspects of the solution. These specifications establish the functional and operational objectives that guided the development of both the physical monitoring station and the supporting digital infrastructure.

Parameter / Requirement	Target value	Type	Rationale / Origin
Monitoring area	Approximately 8 m wide river section	Performance requirement	Defined by the observable area from the fixed installation position
Spatial resolution	Approximately 1 cm or better	Performance requirement	Required to preserve sufficient spatial detail
Image acquisition frequency	One image every 15 minutes	Operational requirement	Required to provide sufficient temporal resolution for continuous monitoring
Operating period	Daylight hours	Operational requirement	Natural illumination is required for optical image processing
Image resolution	Maximum available camera resolution	Performance requirement	Required to preserve the target ground resolution over the monitored area
Data transmission	Automatic wireless transmission of every acquired image	Functional requirement	Required to avoid manual data retrieval and enable future near-real-time monitoring applications
Data management	Remote cloud-based storage	Functional requirement	Required for continuous remote access to monitoring data

Parameter / Requirement	Target value	Type	Rationale / Origin
Power supply	Fully autonomous off-grid operation	Functional requirement	No electrical infrastructure is available at the deployment site
Environmental operation	Long-term operation in a remote Alpine environment	Environmental requirement	Limited accessibility and harsh environmental conditions require reduced maintenance
Structural robustness	Resistance to expected environmental loads with additional safety margin	Performance / safety requirement	Required to ensure reliable long-term deployment under wind and environmental loading
Installation constraints	Installation from the existing riverbank structure	Design constraint	Determined by the physical characteristics and accessibility of the deployment site
Maintenance	Minimal maintenance requirement	Operational requirement	Frequent interventions are impractical due to limited site accessibility
Operational continuity	Continued operation during temporary interruptions of energy availability	System requirement	Required to prevent unnecessary loss of monitoring data

Table 5: System requirements and design specifications.

The following subsections therefore define the principal system specifications that guided the engineering design process and subsequently served as the reference for evaluating the proposed solution.

5.2.2 Functional Specification

The functional specifications define the fundamental capabilities that the proposed monitoring solution is required to provide in order to satisfy the identified project requirements so, rather than prescribing specific technological implementations, they describe the essential functions that the system must perform throughout the monitoring campaign, establishing the reference framework for the subsequent system development. Since these specifications define the overall requirements they are here presented independently of the technological solutions adopted to implement them ensuring that the engineering process remains guided by operational needs rather than by the characteristics of individual technologies, allowing alternative design solutions to be evaluated based on the same functional objectives.

From a functional perspective, the first requirement concerns the periodic acquisition of high-quality images suitable for long-term geological monitoring. The system must periodically observe the monitored river bed according to a predefined acquisition schedule while ensuring the consistency of the collected observations. Maintaining a regular acquisition frequency is essential to guarantee the temporal continuity of the monitoring dataset, allowing geological changes to be identified, compared and mapped over extended

observation periods. This functionality represents the starting point of the monitoring process and directly determines the quality of all subsequent analyses. The monitoring solution shall also ensure that the acquired observations are transferred to a remote infrastructure where it can be securely stored and made available for subsequent processing therefore granting the continuity of the information flow from the field deployment to the digital environment. This capability is particularly important as the remote location of the monitoring station makes local data retrieval impractical and costly both in terms of money and time, requiring the collected observations to remain continuously accessible for monitoring and validation activities.

After data acquisition and transmission the system shall also support the automated extraction of meaningful information from the obtained images, requiring the system to process the collected data using dedicated image analysis techniques able to identify and quantify the features relevant to the objectives of the monitoring campaign presenting the resulting information in a way capable of reducing the required manual analysis from the researchers. Automating this stage of the monitoring process not only reduces the amount of manual interpretation required from researchers but also improves the repeatability and scalability of the analysis, particularly when large image datasets are generated over long monitoring campaigns.

Considered together these functional specifications are able to define the expected behavior of the proposed monitoring solution and establish the reference framework for the implementation of the monitoring station and its successive evaluation.

5.2.3 Operational specifications

While the functional specifications define the capabilities that the monitoring solution must provide, the operational specifications aim at describing the conditions under which these capabilities are expected to be maintained throughout the whole monitoring campaign therefore establishing the performance objectives associated with the continuous operation of the monitoring system.

The primary operational requirement is the ability to sustain long-term autonomous operation in remote alpine environments. The monitoring solution must therefore remain operational for extended periods while also minimizing the maintenance activities that are required in order to keep it running as well as avoid dependence on permanent on-site supervision. This requirement originates from the logistical complexity and economic cost associated with accessing remote alpine sites, where frequent maintenance interventions are neither practical nor desirable during extended monitoring campaigns therefore continuity becomes a fundamental requirement to ensure the effectiveness of efficient long-term geological observations. Long term autonomy is not the only requirement, indeed the monitoring system must guarantee the precise and regular execution of the planned acquisition schedule while preserving the continuity of the generated information throughout the monitoring campaign. It needs to be taken into account that the operational performance of the solution therefore depends not only on the successful execution of each individual acquisition but also on the capability to maintain consistent system availability over time despite changing environmental conditions. Interruptions in the acquisition schedule would reduce the temporal completeness of the monitoring dataset, potentially compromising the interpretation of interest and limiting the reliability of following analyses. Since the monitoring station is intended to operate in locations characterized by limited accessibility, the system is required to also support remote management of the acquired information making it possible to avoid the need for direct human intervention to change such parameters. This operational requirement contributes to reducing maintenance costs and it increases

the overall efficiency of monitoring activities.

Ultimately, these collective operational specifications allowed the establishment of the performance objectives that guided the development of the energy management system, communication infrastructure and overall system integration presented in the following sections.

5.2.4 Environmental specifications

While the functional and operational specifications define what the monitoring solution is expected to accomplish and how it is required to operate, the environmental specifications identify the external conditions within which these objectives must be achieved. Unlike conventional monitoring installations deployed in easily accessible locations, the proposed system is intended for long-term operation in a remote alpine environment, where climatic conditions and limited accessibility become primary engineering constraints rather than secondary considerations making the environmental context one of the principal drivers of the entire design process.

The first environmental constraint concerns the geographical characteristics of the deployment site. The monitoring station is intended to operate in an isolated mountain environment characterized by significant seasonal variability and limited accessibility throughout the year. This requirement directly influenced several design decisions, including the selection of highly reliable components, the definition of an autonomous energy management strategy and the adoption of a modular architecture capable of simplifying future maintenance operations whenever access becomes possible. A second group of environmental specifications originates from the climatic conditions experienced throughout the period monitoring campaign that lasts for years during which the monitoring solution is required to remain operational despite significant seasonal variations in solar irradiance, ambient temperature, precipitation and wind exposure. These environmental factors directly affect the availability of electrical energy, the mechanical loads acting on the supporting structure and the long-term reliability of the installed equipment leading the engineering design to account for these external conditions from the earliest stages of system development ensuring that each subsystem is capable of maintaining its expected performance under realistic operating scenarios rather than ideal laboratory conditions. Environmental conditions also influence the quality of the monitoring data itself. Variations in illumination, shadows, snow cover and atmospheric visibility may affect image quality and consequently the reliability of the subsequent image analysis procedures. For this reason, environmental variability cannot be considered solely as a constraint affecting the physical infrastructure but must also be taken into account throughout the design of the digital processing workflow leading to the necessity to consider the interaction between environmental conditions and data processing during the development of the monitoring solution.

Collectively, these environmental specifications define the external constraints within which the proposed monitoring solution is expected to operate as they establish the reference conditions adopted throughout the engineering design process and provide the basis for the structural analyses, energy system sizing and subsystem selection presented in the following sections.

These specifications provide the common engineering basis for the design of the subsequent subsystems. The imaging subsystem is required to satisfy the spatial and temporal monitoring requirements, while the energy subsystem is sized according to the resulting operational profile and electrical demand. Similarly, the mechanical infrastructure must support the selected components while ensuring reliable operation under the environmental and installation constraints of the deployment site. The following sections therefore do not introduce independent requirements, but describe how the system-level specifications defined in this

section were translated into subsystem design choices and subsequently verified.

5.2.5 Engineering design framework

The functional, operational and environmental specifications presented in this section establish the engineering framework adopted throughout the development of the proposed monitoring solution defining a coherent set of constraints and performance targets.

The proposed monitoring solution is composed of several interconnected subsystems, each designed to satisfy a specific subset of the identified system specifications while contributing to the overall performance of the monitoring solution. Consequently, the subsequent chapters examine the engineering design of the imaging, structural, electrical, energy and communication subsystems together with the image processing pipeline, illustrating how each design decision contributes to the implementation of the complete monitoring solution.

5.3 Imaging Subsystem

5.3.1 Design Drivers

The system specifications presented in Section 5.2 define the performance objectives of the monitoring solution and from the perspective of the imaging subsystem these specifications translate into a set of design drivers that directly influenced the selection and configuration of the adopted imaging technology. The primary design driver concerned the simultaneous achievement of two conflicting objectives: monitoring an approximately 8 m wide riverbed while maintaining a ground resolution close to 1 cm, allowing the subsequent Machine Learning algorithms to reliably identify the geomorphological features of interest. Achieving these two objectives simultaneously required the adoption of the widest field of view compatible with the required spatial resolution, making optical performance one of the primary selection criteria. Furthermore a temporal sampling interval of 15 minutes, corresponding to 48 images per day, was selected as the reference operating condition. This acquisition frequency represented a compromise between the temporal resolution required for long-term monitoring, the limitations imposed by energy availability, communication bandwidth and storage capacity together with the restriction of the image acquisition period during the day for a flexible acquisition interval allowing the system to exploit natural illumination while simultaneously reducing unnecessary energy consumption during nighttime hours. The last relevant requirements influencing the subsystem design refer to the amount of data to be communicated periodically. An average image size of 5-7 MB is estimated thus the monitoring station generates approximately 250-335 MB/day and approximately 90-100 GB/year, making automatic wireless transmission and cloud-based data management essential elements of the proposed monitoring architecture. The final configuration therefore results not only from the optimization of the imaging subsystem itself but also from the interactions with the energy management, communication and image processing subsystems.

5.3.2 Camera selection and tradeoffs analysis

The selection of the imaging device therefore consisted in evaluating the available technologies against the system specifications previously identified consequently leading to the necessity of assessing characteristics beyond image quality alone and the consideration of the combined effects of optical performance, energy consumption, communication capabilities, operational autonomy, environmental robustness and ease of system integration. Among the identified design drivers the simultaneous achievement of the required

ground resolution and field of view represented one of the most critical selection criteria given the fixed installation position imposed by the characteristics of the monitoring site. These two requirements had to be simultaneously achieved by selecting an imaging system capable of combining a sufficiently wide field of view with adequate image resolution and as such, solutions characterized by narrower viewing angles would have required either increasing the installation distance, which was not feasible, or sacrificing spatial resolution over the monitored area, reducing the effectiveness of the image analysis stage at identifying smaller details and water courses.

Besides optical performance, autonomous operation represented an equally important design criterion. The acquisition strategy defined in the previous section also imposed strong constraints on hardware selection as to the acquisition frequency during daylight hours, followed by the wireless transmission of every single acquired image to the remote cloud infrastructure. Such an operating strategy gives birth to constraints in terms of energy consumption since image acquisition, data communication and system management must all be sustained exclusively through the autonomous power supply discussed in the following sections and therefore the selected camera has to minimize power demand while simultaneously integrating wireless communication capabilities. Conventional industrial cameras or DSLR-based solutions generally require external computing platforms, additional communication hardware and continuous power supply, significantly increasing both system complexity and overall energy consumption also due to the presence of multiple distinct modules with their respective electrical inefficiencies. Similarly, Raspberry Pi-based configurations, although highly flexible, require the integration of several independent hardware and software components, resulting in greater design complexity and higher overall power requirements once modules for communication, processing and energy management are considered. The main imaging alternatives were therefore evaluated against the requirements identified in Section 5.2. The comparison summarized in Fig 5 shows that no alternative simultaneously provides the required field of view, spatial resolution, autonomous operation, integrated communication and low system complexity without requiring additional external hardware. On this basis, the Tikee 4X was selected as the most suitable commercially available solution for the proposed monitoring architecture. Infrared imaging solutions were also excluded since the additional information provided was not justified by the considerable increase in energy consumption as such solutions were characterized by increases of up to 1000% compared to traditional cameras. On the other hand, submerged camera configurations were considered incompatible with the operational environment due to the high probability of damage caused by transported debris, sediment accumulation, the limited accessibility of the monitoring site and the possibility of changes in the river morphology throughout the year not granting the certainty of being submerged, and therefore acquiring data, for the whole acquisition period.

The engineering assessment discussed above identified the Tikee 4X, shown in Fig 6 as the only candidate capable of satisfying all the critical system specifications while minimizing energy consumption compatible with realistic production possibilities and avoiding further system complications. The device integrates a dual 12 MP imaging system, providing a stitched panoramic image with a horizontal field of view of approximately 220° and a vertical field of view of approximately 90°, allowing the required riverbed area to be monitored while maintaining an estimated ground resolution of approximately 7.7 mm at the center of the monitored section. The camera also incorporates native LTE connectivity, programmable acquisition schedules, 128 GB of internal storage, a 25 600 mAh internal battery and an integrated 4 W photovoltaic panel within a single weatherproof enclosure weighing approximately 1.6 kg, however, under the selected acquisition schedule, the internal battery alone is capable of sustaining approximately one week of operation, providing an additional level of robustness in the event of temporary failures of the external power system but not granting alone long enough autonomy for problem identification and consequent on

CRITERION	Weight	Raspberry pi	Industrial Camera	DSLR Camera	Tikee4X
Ground resolution	High	✓	✓	✓	✓
Wireless connection	High	△	✗	✗	✓
Horizontal FOV	High	△	△	✗	✓
Power consumption	High	△	✗	✗	✓
Reliability	Medium	△	✓	△	✓
Ease of deployment	Medium	✗	✗	✗	✓
Internal battery	Low	✗	✗	✗	✓

Figure 5: Comparison of the evaluated imaging solutions against the principal system requirements. Symbols indicate whether each requirement is fully satisfied (V), partially satisfied (△) or not satisfied (X).

site intervention.

Overall, the selected imaging device satisfied the optical, energetic and operational specifications simultaneously while significantly simplifying the integration of the overall monitoring station. The high level of integration offered by the selected camera also reduced the complexity of the overall monitoring station by eliminating the need for dedicated acquisition hardware and simplifying both electrical integration and system deployment ensuring that the imaging subsystem fulfilled the required performance without introducing unnecessary complexity or compromising the long-term reliability of the deployment.

5.3.3 Camera configurations and Operating parameters

Once the imaging device had been selected, its operating parameters, recalled in Fig 6 were configured to satisfy the monitoring objectives defined in the previous sections while maintaining the compatibility with the energy and communication constraints of the overall monitoring system. Although the Tikee 4X provides a large number of configurable acquisition options, only those directly affecting image quality, temporal sampling, communication strategy and power consumption were modified aiming to the objective of maximizing the amount of useful information acquired while ensuring long-term autonomous operation. The acquisition schedule was configured to automatically adapt to the natural daylight conditions at the deployment site rather than defining a fixed operating interval throughout the year, this way image acquisition was synchronized with local sunrise and sunset, allowing the camera to operate exclusively under natural illumination while also avoiding unnecessary acquisitions during nighttime and automatically compensating for the progressive variation in daylight duration between summer and winter. An acquisition interval of 15 minutes was adopted, representing the compromise previously identified between temporal resolution, energy availability and communication requirements. Under average daylight conditions this configuration results in 48 acquired images per day while maintaining sufficient temporal continuity to observe the evolution of the monitored riverbed without generating unnecessary redundancy. The camera was configured to acquire images at its maximum available resolution of 4032×3024 pixels in JPEG format as among the alternative formats that could potentially preserve additional image information, JPEG represents the only acquisition format compatible with the automatic wireless transmission required by the monitoring strategy as per manufacturer limit. Maintaining the maximum image resolution was considered essential to preserve the target ground resolution of approximately 7.7 mm over the monitored



Figure 6: Installed Tikee 4X camera showing the final field deployment configuration.

area allowing the subsequent Machine Learning algorithms to reliably identify small geomorphological features and narrow water channels. The panoramic image generated by the camera is obtained through the combination of the two integrated imaging sensors, providing the wide field of view required to monitor the complete river section preserving adequate spatial detail across the entire scene. Lastly the automatic exposure and automatic white balance were maintained throughout the monitoring campaign in order to continuously adapt the acquisition parameters to the changing illumination conditions without requiring manual intervention or periodic recalibration. The resulting image quality and acquisition geometry were also selected to provide suitable input data for the image-analysis pipeline described in Section 5.6.

Parameter	Selected value	Design rationale
Ground resolution	Approximately 7.7 mm	Provides sufficient spatial detail to identify small geomorphological features and narrow water channels.
Field of view (HFOV $\times$ VFOV)	$220^\circ \times 90^\circ$	Allows coverage of the complete monitored river section.
Acquisition interval	15 min	Represents the selected compromise between temporal resolution, energy consumption and communication requirements.

Parameter	Selected value	Design rationale
Operating period	Sunrise–sunset	Uses natural illumination, avoids unnecessary nighttime acquisitions and automatically adapts the operating period to seasonal variations in daylight duration.
Image format	JPEG	Provides a suitable compromise between image quality and file size.
Image resolution	4032×3024 pixels	Maximum available resolution to preserve the target ground resolution.
Image upload	Every acquired image	Ensures continuous remote availability of the monitoring data and supports future near-real-time monitoring or early-warning applications.
Exposure	Automatic	Adapts the acquisition parameters to changing illumination conditions without requiring manual intervention or periodic recalibration.
White balance	Automatic	Maintains consistent image acquisition under changing illumination conditions and seasonal variations in natural light.
Always-On mode	Disabled	Allows the camera electronics to remain in a low-power state between acquisition events, reducing the average energy consumption.

Table 6: Selected camera operating parameters and corresponding engineering rationale.

The communication strategy was configured to automatically upload every acquired image immediately after acquisition through the integrated LTE connection to the Tikee Cloud platform. While LPWAN technologies such as LoRa are particularly advantageous for low-data-rate sensor networks, the present application requires the transmission of several megabytes for each acquired image. Direct cellular connectivity was therefore considered more appropriate for the required data throughput while avoiding the additional infrastructure associated with an intermediate gateway. Although periodic batch transmission would reduce communication activity and potentially decrease energy consumption, immediate transmission was preferred because it guarantees continuous remote availability of the monitoring data and preserves the possibility of future real-time applications, including the implementation of automatic early-warning procedures. The camera was additionally configured to operate without the Always-On mode allowing the internal electronics to remain in a low-power state between consecutive acquisitions that, since the monitoring station spends the majority of its operational lifetime waiting for the next scheduled acquisition, allows considerable reductions in the average energy consumption contributing directly to the extension of the overall system autonomy. A complete state-machine diagram can be observed in Fig 7.

Overall, the adopted operating configuration represents the engineering compromise between image quality, temporal resolution, communication efficiency and energy consumption required by the monitoring

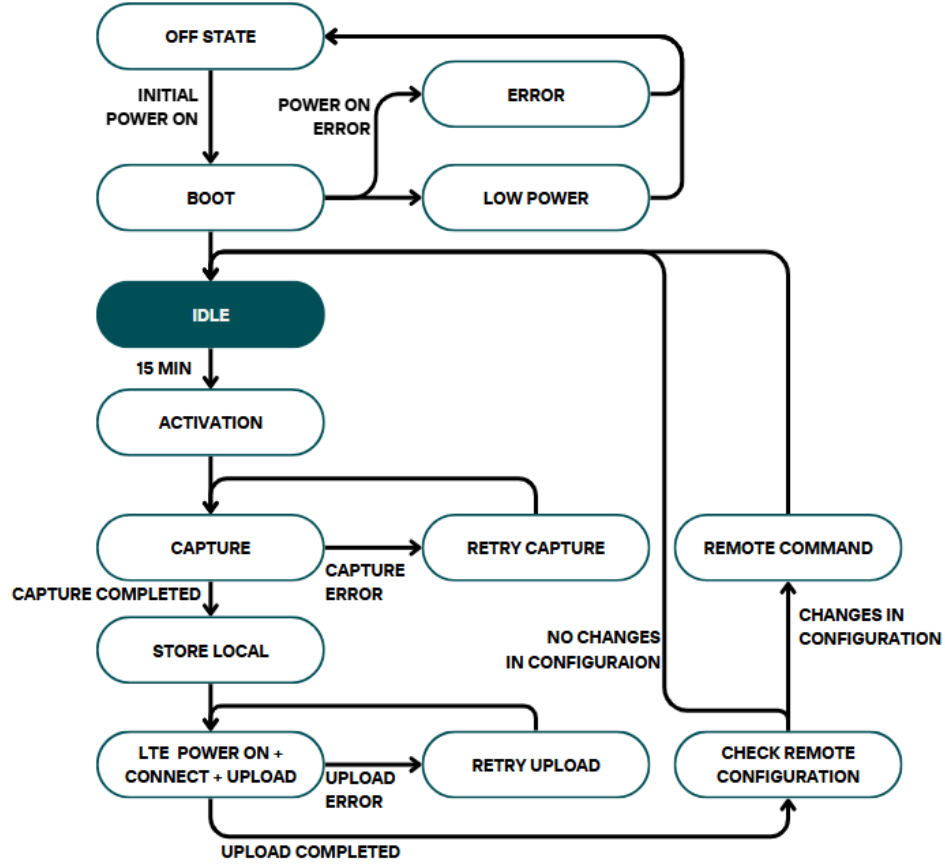


Figure 7: Operational state machine of the Tikee 4X during the acquisition cycle.

campaign as, rather than optimizing a single acquisition parameter independently, the final configuration was defined to ensure that all selected settings remain consistent with the overall objectives of the complete monitoring system while supporting reliable long-term autonomous operation. The resulting imaging subsystem provides the complete acquisition capability required by the proposed monitoring solution by combining the selected hardware platform with an operating configuration specifically tailored to the project requirements. The adopted design ensures that the monitoring station is capable of autonomously acquiring, storing and transmitting high-quality observations while remaining compatible with the energetic and operational constraints imposed by long-term deployment in a remote alpine environment. Having established the imaging subsystem, the following section addresses the design of the energy subsystem required to sustain its continuous autonomous operation.

5.4 Energetic subsystem

5.4.1 Design Drivers

The system specifications presented in Section 5.2 establish the performance objectives that govern the design of the energy subsystem. From the perspective of the power supply architecture these objectives translate into a set of design drivers aimed at guaranteeing continuous autonomous operation while at the same time ensuring compatibility with both the environmental conditions of the deployment site and the operational profile of the imaging subsystem.

The primary design driver concerns the ability to sustain long-term operation without relying on any external electrical infrastructure since the monitoring station is intended to remain deployed for several years in a remote alpine environment characterized by limited accessibility thus the energy subsystem must continuously satisfy the electrical demand of the complete monitoring station and minimize maintenance interventions, preserving operational continuity throughout the monitoring campaign. The direct consequence is that the available energy cannot be dimensioned solely according to average operating conditions but must also account for the significant seasonal variability in solar resource availability discussed in the previous chapter. The second design driver originates from the operating strategy adopted for the imaging subsystem. The acquisition schedule defined in Section 5.3 consists of one acquisition event approximately every 15 minutes during daylight hours, with each image being automatically transmitted through the integrated LTE communication system immediately after acquisition. Based on this operating profile, the reference daily energy demand used for subsystem sizing was estimated at approximately 6.16 Wh/day. As this value could not be obtained directly from manufacturer measurements, it represents an engineering estimate derived from the available data and adapted to the acquisition and transmission scenario defined for the proposed monitoring system. The adopted estimation procedure is presented in the following section together with the corresponding subsystem sizing methodology.

Ultimately, the electrical architecture of the monitoring station introduces additional design constraints that must be considered during the development of the energy subsystem. While the autonomous power system operates on a 12 V DC line, the selected imaging device requires a regulated 5 V supply making voltage conversion an essential part of the overall design and as such conversion efficiency together with associated power losses must also be considered during the sizing process to ensure that the available energy remains sufficient under all expected operating conditions. Taken together these design drivers establish the engineering framework for the development of the energy subsystem, providing the basis for the energy budget estimation, component sizing and architecture selection presented in the following sections.

5.4.2 Energy system sizing methodology

The first step required is the estimation of the energy demand to sustain the desired acquisition schedule shown in details in Fig 8. Using data contained in the technical documentation available a use case with the same acquisition frequency, daily acquisition interval, median image quality and single image upload present an estimated power consumption equal to 4.14 Wh/day. From this data the energy consumption per event can be retrieved and adapting the energy consumption demanded by the upload process so that the image transferred maintains the maximum allowed quality the daily energy demand required for the implementation of the proposed solution can be estimated at 6.16 Wh/day. Following the estimation of the energy demand, the design of the power supply subsystem consisted in identifying the photovoltaic generator and battery capacity required to guarantee uninterrupted operation throughout the entire

monitoring campaign. For this analysis rather than relying exclusively on nominal operating conditions the sizing procedure was developed through a progressive engineering approach including the assessment of the available solar resource, the preliminary sizing of the photovoltaic as well as storage systems and the verification of the proposed solution through sensitivity analyses and long-term dynamic simulations allowing the robustness of the final design to be evaluated under realistic environmental conditions rather than ideal operating scenarios. The sizing procedure was based on the estimated average daily energy consumption equal to approximately 6.16 Wh/day. Firstly, since the imaging subsystem operates at 5 V whereas the photovoltaic system and the battery operate at 12 V, the energy balance also accounts for the DC/DC conversion stage, for which a conservative efficiency equal to 90% was assumed throughout the design process and the photovoltaic generator was required to produce sufficient energy not only to satisfy the daily electrical demand of the monitoring station but also to compensate for the conversion losses associated with the voltage regulation stage.

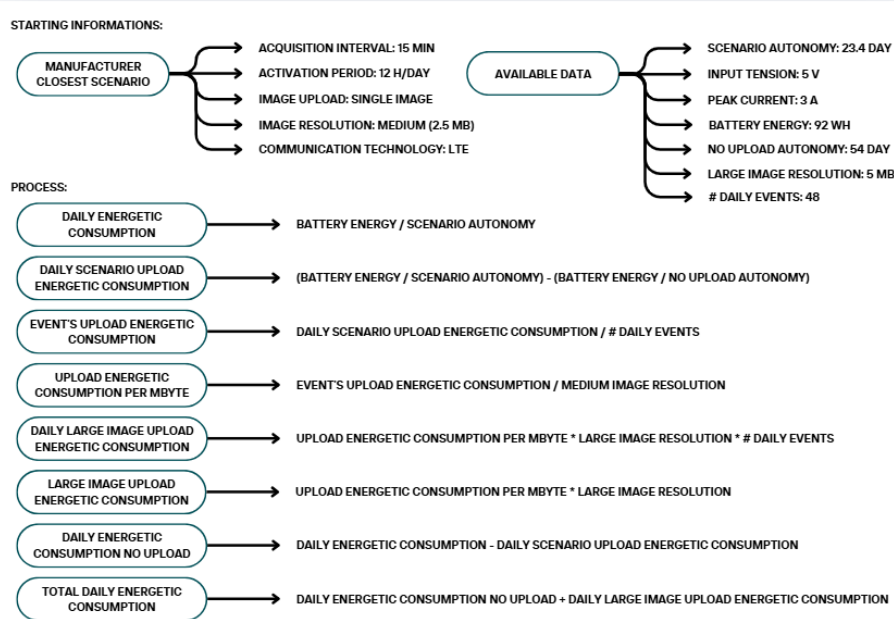


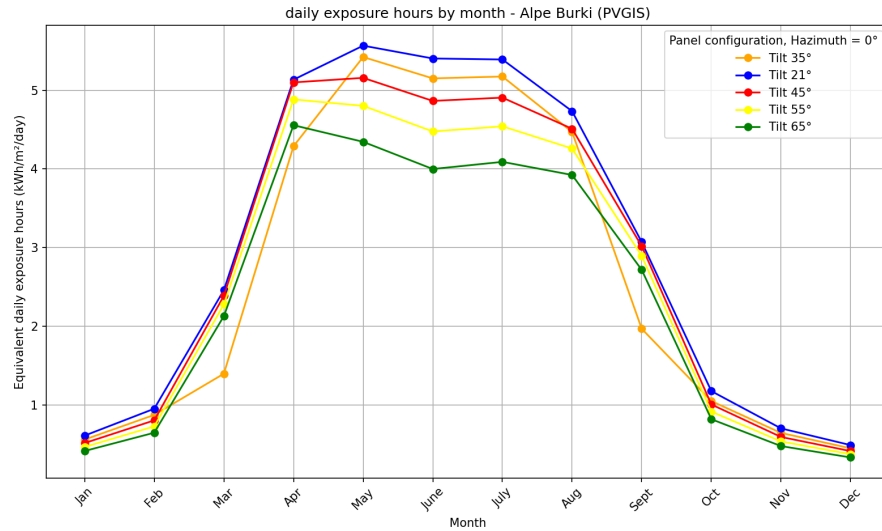
Figure 8: Procedure adopted to estimate the daily energy consumption of the Tikee 4X under the selected operating conditions.

Before selecting the photovoltaic module the available solar resource at the deployment site was analysed. Since the monitoring station is installed within a narrow alpine valley the actual solar availability cannot be accurately estimated considering astronomical irradiance alone because the surrounding mountains significantly reduce the effective sunlight duration, particularly during the winter months and to account for this aspect the photovoltaic resource assessment was performed using PVGIS, a software that allows the local topography and horizon profile surrounding the installation site to be included in the calculation of the available solar irradiation. Different panel inclination angles were also evaluated in order to identify the configuration providing the most favorable compromise between annual energy production and winter performance, including avoidance of snow pile up over the generator. Particular attention was dedicated to the minimum monthly Peak Sun Hours (PSH), considered the reference standard to

compare different alternatives as the energy subsystem must remain operational even during the least favorable period of the year. The comparison between the analysed inclinations is reported in Fig 9, from which an inclination of approximately 55° with respect to the horizontal was selected although neighboring inclinations provide comparable annual irradiation values as the chosen one offers improved winter performance while simultaneously reducing the probability of snow accumulation on the photovoltaic surface, therefore increasing the long-term reliability of the installation.

Once the available solar resource had been characterized, a preliminary sizing of the photovoltaic generator and battery storage system was performed based on the estimated daily consumption, the worst-case winter solar availability and the assumed conversion efficiency, identifying the minimum required photovoltaic power. The resulting analysis led to the adoption of a 20 W photovoltaic panel coupled with a 12 V, 12.8 Ah $LiFePO_4$ battery, corresponding to an overall storage capacity of approximately 144 Wh. This preliminary configuration guarantees that the average daily energy production exceeds the estimated electrical demand under nominal operating conditions throughout the year while simultaneously providing sufficient energy storage to sustain continuous operation during periods of reduced solar availability even in case of adverse events related variations to the production rates for prolonged period of time, sufficient to allow identification and physical intervention to take place. Considering the combined solar power generation of both the external power supply and the integrated panel in the Tikee4 device as well as the combined electrical reservoir composed of the external $LiFePO_4$ battery and the one integrated in the camera, a totality of 24W solar energy production and 236 Wh energy storage is achieved. It is important to differentiate these electrical parameters from those described before, amounting to 20W of energetic production and 144 Wh of storage capacity as those refer just to the external power subsystem adopted while those presented above correspond to the integrated energetic resources combining both the external system and the characteristic of the camera itself. Consequently, a minimal production rate of 7.2 Wh/day is achieved throughout the year during the winter months making the production rate in the worst considered conditions higher than the daily demand even considering the previously described efficiency and an autonomy of the system to operate for 38 continuous days is achieved even in case of complete production shutdown, allowing maintenance crew to detect the problem, organize the opportune intervention strategy and circumvent eventual problems related to temporary adverse conditions that may take place, especially during winter.

Although the preliminary sizing satisfies the nominal energy balance, the actual operating conditions of the monitoring station are characterized by unavoidable uncertainties associated with both energy production and energy consumption. Variations in weather conditions, component aging, conversion efficiencies and communication activity may all influence the effective energy balance throughout the monitoring campaign consequently introducing the need for the robustness of the proposed design to be evaluated through a comprehensive sensitivity analysis considering simultaneous variations of both the daily energy demand and available photovoltaic production. The analysis considered variations in the monitoring station energy consumption ranging from -30% to +50% with respect to the nominal estimate together with photovoltaic production variations between -50% and +30% of the expected values and for every combination the corresponding energy margin, defined as the ratio between daily energy production and daily energy consumption, was calculated. The obtained sensitivity matrix is presented in Fig 10 showing the minimum energy margin obtained throughout the year. Under nominal operating conditions the proposed configuration provides an energy margin equal to approximately 1.052, indicating that the minimum daily photovoltaic production exceeds the expected electrical demand. As expected, the energy margin progressively decreases as the available solar resource is reduced or the system consumption increases, allowing the operating limits of the proposed configuration to be identified before deployment.



Daily exposure hours by month - equivalent PSH

Month	Tilt 21°	Tilt 35°	Tilt 45°	Tilt 55°	Tilt 65°
January	0,6105	0,5602	0,5158	0,4665	0,4151
February	0,9521	1,8737	0,8044	0,7276	0,6474
March	2,2460	4,1397	2,2387	2,2278	2,2129
April	5,5134	5,4288	5,5100	4,4883	4,4556
May	5,5567	5,5422	5,5157	4,4802	4,4345
June	5,5404	5,5151	4,4864	4,4476	3,3999
July	4,5391	5,5174	4,4906	4,4542	4,4090
August	3,4734	4,4470	4,4510	4,4259	3,3923
September	1,3071	1,1969	3,3018	2,2894	2,2717
October	0,1176	1,1050	1,1006	0,9155	0,8192
November	0,7026	0,6448	0,5936	0,5369	0,4777
December	0,4898	0,4495	0,4138	0,3743	0,3330

Figure 9: Monthly Peak Sun Hours (PSH) obtained from PVGIS for the analysed photovoltaic panel inclinations: graph (top), numerical equivalent (bottom).

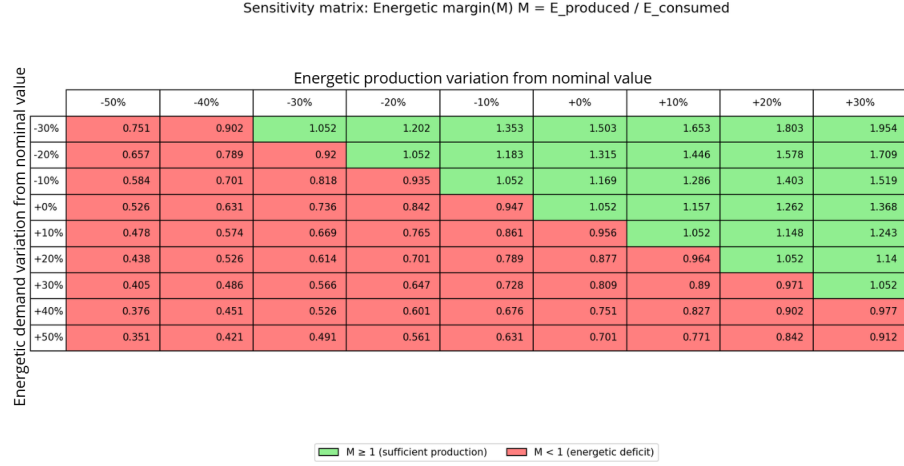


Figure 10: Sensitivity analysis of the annual minimum energy margin as a function of photovoltaic energy production and system energy consumption.

While the previous analysis provides a static evaluation of the minimum energy balance, the long-term behavior of the monitoring station is better described by the cumulative evolution of the battery state of charge throughout the year. For this reason, a dynamic simulation was performed considering the monthly solar irradiation obtained from PVGIS together with the estimated daily electrical demand evaluating the temporal evolution of the battery charge over an entire annual operating cycle and therefore accounting for the seasonal variability of the available solar resource rather than considering isolated operating conditions. The simulation performed under nominal conditions shows that the selected energy subsystem is capable of maintaining the battery at full charge throughout the entire year, further proving that the available photovoltaic generation remains sufficient to compensate for the daily electrical demand even during the least favorable months. In order to verify the robustness of this result, the dynamic simulation was repeated considering the same production and consumption variations adopted in the sensitivity analysis and the corresponding minimum annual battery state of charge obtained for the analysed scenarios, reported in Fig 11, allowing the combinations of environmental conditions and electrical demand that may compromise long-term autonomous operation to be identified. The obtained results are more informative compared to the minimum energetic margin as it shows under which conditions the system is able to operate even if not always at full charge and, on the other hand, in which the system will collapse showing that even under significant worsening conditions the monitoring solution will be able to operate as intended.

Overall, the adopted sizing methodology combines analytical calculations, realistic solar resource assessment and numerical verification in order to validate the proposed energy subsystem under both nominal and non-ideal operating conditions. Rather than relying on conservative component oversizing alone, the proposed approach demonstrates that the selected external 20 W photovoltaic module and 144 Wh battery provide sufficient energy availability to sustain continuous operation of the monitoring station while maintaining adequate robustness against realistic variations in both environmental conditions and system energy demand.

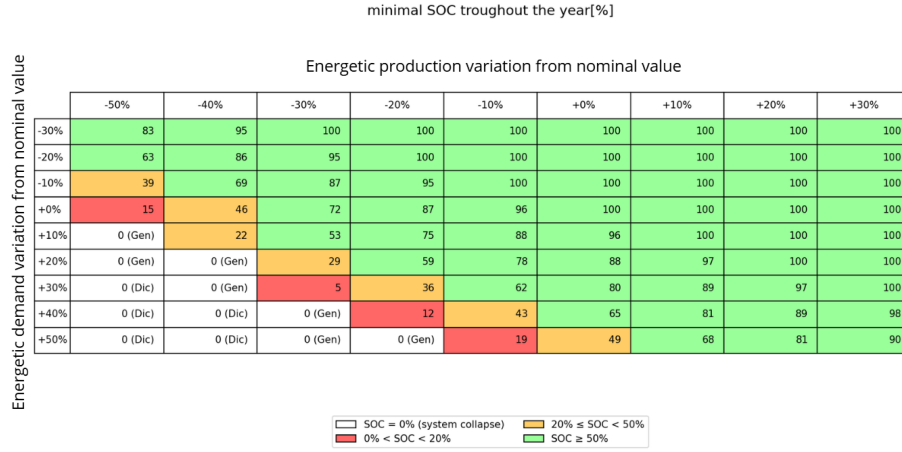


Figure 11: Minimum annual battery State of Charge obtained for the analysed production and consumption scenarios.

5.4.3 Hardware Implementation

Once the sizing methodology had identified the required characteristics of the power supply system, the energy subsystem was implemented through the integration of commercially available components selected to satisfy the electrical, environmental and operational requirements previously discussed. Rather than focusing exclusively on the nominal electrical performance of each individual component, the implementation aimed at obtaining a compact, reliable and easily maintainable subsystem capable of supporting long-term autonomous operation under the environmental conditions expected at the deployment site, hence the final architecture consists of a photovoltaic generator, a $LiFePO_4$ battery storage system, a solar charge controller, a DC/DC voltage converter and the associated protection and distribution circuitry, as illustrated in Fig 12.

The photovoltaic generator is based on a 20 W monocrystalline solar panel, selected according to the sizing procedure presented in the previous section that provides a maximum power output of 20 W with a nominal operating voltage (V_{mp}) of 12 V and a maximum operating current ($I_{mpulsive}$) of 1.11 A, providing an adequate compatibility with the selected charge controller while also ensuring sufficient power generation during the winter months identified as the most critical operating condition. The module itself measures $410 \times 345 \times 25$ mm, has a mass of approximately 1.8 kg and operates over a temperature range between -40°C and $+85^\circ\text{C}$ while also incorporating an IP67 protected junction box and an integrated bypass diode to improve reliability under partial shading conditions avoiding inversion of polarity on the line directed to the charge controller. Thanks to its relatively compact dimensions it also simplifies the mechanical integration of the monitoring station without introducing excessive structural loads on the supporting frame. Energy storage is provided by a 12 V, 12.8 Ah $LiFePO_4$ battery, allowing the usage of an overall external storage capacity of approximately 144 Wh. Compared with conventional lead-acid technologies, lithium iron phosphate batteries provide higher energy density, reduced weight, improved cycle life and greater efficiency making them particularly suitable for autonomous outdoor monitoring systems where maintenance activities are intentionally minimized. The selected battery has a mass of approximately 0.5 kg and incorporates an integrated Battery Management System (BMS) responsible

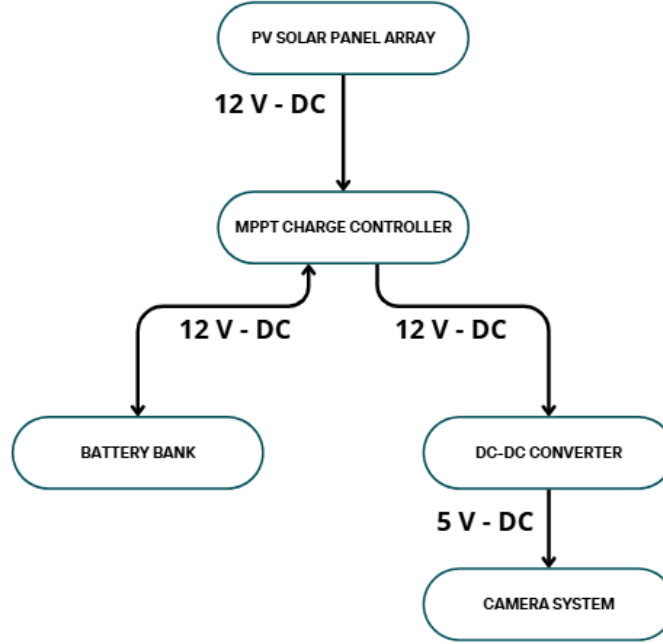


Figure 12: Functional architecture of the energy subsystem, including the photovoltaic panel, MPPT charge controller, battery, DC/DC converter and imaging device.

for protecting the battery against abnormal operating conditions since $LiFePO_4$ batteries should not be charged below 0°C , furthermore an additional Victron Smart Temperature Sensor was integrated into the system that through Bluetooth communication with the charge controller is able to continuously monitor the battery temperature and automatically inhibits the charging process whenever the operating temperature falls below the recommended threshold, preventing irreversible battery degradation under extremely harsh conditions. The sensor is provided electrical energy through a direct connection with the battery itself however its consumption has not been taken into account during the system sizing presented above as the nominal power consumption is listed in the order of $\mu\text{Wh/day}$, being considered not comparable with the daily power demand of the image acquisition system. The management of the photovoltaic generation is performed by a Victron Energy MPPT solar charge controller, which represents the central element of the electrical architecture thanks to its ability to redirect current flows depending on the operational necessities of the complete system further enabling automatic temperature-compensated battery management thanks to its Bluetooth connection with the temperature sensor without requiring additional control electronics and thereby improving both system reliability and implementation simplicity. Since the monitoring camera requires a regulated 5 V supply whereas the energy storage system operates at 12 V, the electrical architecture also incorporates a 15 W DC/DC buck converter capable of converting an input voltage between 9.2 V and 18 V into a stable 5 V output with a maximum current of 3 A, the same peak current required by the camera therefore providing the perfect electrical interface between

the battery and the imaging subsystem maintaining a stable supply voltage independently of the battery operating conditions.

The electrical components of the power supply subsystem are housed inside a dedicated IP65 electrical enclosure. Besides providing mechanical support for the electrical architecture, the enclosure protects the installed equipment from precipitation, dust and accidental external contact, allowing the monitoring station to operate reliably under the harsh environmental conditions expected. The enclosure has overall dimensions of $400 \times 300 \times 200$ mm and is manufactured from carbon steel in order to provide adequate mechanical strength while maintaining a compact footprint. All external cable entries are realized through IP68 cable glands equipped with integrated sealing gaskets, ensuring the integrity of the enclosure. Since the monitoring station is intended to remain operational throughout the entire year, particular attention was also dedicated to the internal environmental conditions of the enclosure thus a 19 mm thermal rubber insulation layer was installed on the internal walls in order to prevent the dissipation of heat produced by the electrical devices and limiting the influence of the external climate on the operating conditions of the power supply subsystem reducing consequently the possibilities of reaching sub-zero temperatures that result in the interruption of the recharge process of the battery. During the summer months, the insulation helps maintain the temperature inside the electrical enclosure below the external ambient temperature, thereby reducing the likelihood of damage associated with self-heating, however, such damage can be considered unlikely, as the operating temperature range specified by the component manufacturers extends up to 70 °C. The internal layout was designed to provide adequate separation between the photovoltaic regulator, battery, DC/DC converter and protection devices while simultaneously maintaining sufficient accessibility for inspection and maintenance activities whenever field interventions become necessary as shown in Fig 13.

Particular attention was also dedicated to the reliability of the electrical installation as independent fuse protection was introduced on both the photovoltaic input line and the battery supply line in order to protect the system against accidental short circuits and wiring faults. Furthermore, all power connections were realized using 4 mm² electrical cables, selected to minimize Joule's effect related power losses along the relatively short electrical connections while simultaneously providing improved mechanical robustness and insulation suitable for prolonged outdoor operation since the cables are present both inside and outside the electrical case. Other non-electrical safety measures have been implemented as well as physical shielding has been integrated, providing all external cables with the opportune tear resistant sleeves avoiding the degradation of the insulating material around the conductive element of the cables.

Overall, the implemented energy subsystem directly reflects the sizing methodology previously presented as each component was selected not only to satisfy its individual electrical function but also to ensure compatibility with the remaining subsystems, resulting in an integrated architecture capable of providing reliable long-term power supply while maintaining the simplicity and robustness required for autonomous deployment in remote alpine environments.

5.5 Mechanical Subsystem

5.5.1 Mechanical design drivers

The mechanical subsystem is responsible for providing the physical support required by the monitoring station while ensuring that all the integrated components maintain their intended functionality throughout the entire monitoring campaign. Unlike conventional monitoring installations operating under controlled conditions, the proposed system is intended for long-term deployment in a remote alpine environment,



Figure 13: Internal layout of the electrical enclosure showing the arrangement of the main components and cable routing after final assembly.

where the supporting structure is continuously exposed to environmental actions capable of affecting both structural integrity and monitoring performance, hence the mechanical design was developed not only to support the installed equipment but also to preserve the operating conditions required by the imaging, energy and communication subsystems. The primary design requirement concerned the structural stability of the monitoring station since the camera continuously observes the same river section over periods potentially lasting several years thus any significant displacement or rotation of the supporting structure would directly affect the repeatability of the acquired images and consequently reduce the reliability of the image analysis procedures based on them. The mechanical subsystem was therefore required to provide sufficient stiffness to limit structural deformations under the expected environmental loads while simultaneously guaranteeing an adequate safety margin against material failure. A second design driver originated from the environmental conditions characterizing the installation site. Wind loading represents the most significant external action acting on the monitoring station due to the exposed installation position and the presence of multiple components exposing different projected areas including the photovoltaic panel, the electrical enclosure and the imaging device. Furthermore, the supporting structure is expected to remain operational under large seasonal temperature variations and prolonged exposure to precipitation and humidity, requiring the selection of corrosion-resistant materials and protective solutions capable of ensuring long-term reliability with minimal maintenance requirements. Mechanical integration also represented an important design objective as the camera, photovoltaic panel and electrical enclosure needed to be

mounted on a common supporting structure while maintaining appropriate orientation, accessibility and weight distribution hence the final configuration had to ensure not only adequate structural performance but also facilitate installation, maintenance and future subsystem replacement whenever required. These design drivers collectively established the engineering framework adopted for the structural development of the monitoring station, whose configuration and verification are presented in the following sections.

5.5.2 Structural design and verification

Based on the mechanical design drivers discussed in the previous section, the monitoring station was developed around a vertical steel supporting structure anchored to an existing reinforced concrete platform located along the riverbank that consists of a 2 m high steel pole with an external diameter of 50 mm, providing the supporting element for all the components required by the proposed solution being secured to the concrete foundation through a bolted base connection, while three steel guy wires provide additional lateral stiffness and improve the global structural behavior under environmental loading conditions. The final solution's model used for simulations is shown in Fig 14. Rather than designing a completely new supporting infrastructure the proposed solution takes advantage of the existing civil structure available at the monitoring site, reducing installation complexity while ensuring adequate elevation of the imaging system above the monitored river section. The spatial arrangement of the installed components was defined not only to satisfy their individual operational requirements but also to optimize the overall mechanical behaviour of the monitoring station. The imaging device is positioned in the higher half of the supporting pole through an adjustable ball-head mounting bracket, allowing accurate orientation towards the monitored riverbed while preserving the required field of view. The photovoltaic panel on the other hand is installed at the highest possible point on the pole using a dedicated support manufactured from 20×20 mm stainless-steel extruded profiles in order to maintain the inclination angle of 55° previously identified during the energy subsystem design while also being able to withstand wind loads applied to the panel whose vertical projected area is equal to 11.59 cm^2 and fixed to the pole through 3 U-bolt clamps characterized by a diameter of 8 mm distributing the transmitted loads over a larger contact area and reducing local stresses within the supporting structure. The electrical enclosure is mounted in the lower portion of the pole through two of the same stainless-steel U-bolt clamps used for the panel allowing the arrangement to also lower the overall center of mass of the monitoring station while facilitating maintenance operations and improving accessibility to the electrical components.

Following the definition of the structural configuration, numerical analyses were performed to verify the capability of the supporting structure to withstand the expected environmental loads throughout the monitoring campaign. The structural behavior was investigated using the Finite Element Method (FEM) implemented in SolidWorks Simulation and to simplify the numerical model while preserving the global mechanical response of the system detailed geometries of bolts, screws and threaded connections were neglected being replaced by idealized bonded or rigid connections between the assembled components. The supporting pole was modeled according to its actual geometry while the anchoring system reproduced the constraints imposed by the reinforced concrete support. Such a modeling approach considerably reduced computational complexity without significantly affecting the accuracy of the global structural response which represents the primary objective of the analyses. Static structural analyses were carried out by combining the self-weight of the monitoring station with wind loads acting from different directions. Wind applied loads have been calculated based on data from EMA database, containing data about wind speed at 10 m above ground, hence the necessity to correct such observation using data from ground based meteorological stations even though the latter are situated in more exposed positions compared to the

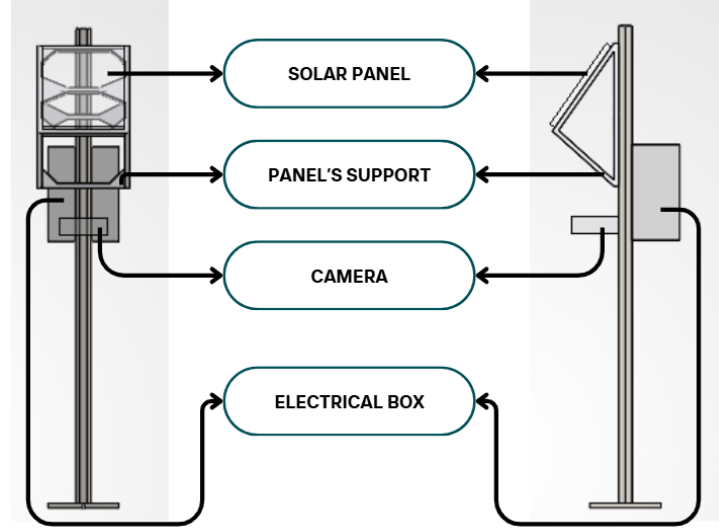


Figure 14: CAD model of the complete monitoring station showing the mechanical integration of the main subsystems.

proposed solution allowing the hypothesis that starting data include an undefined but present safety margin. Although the maximum historical wind speed recorded in the region around the deployment site is approximately 18 m/s, additional simulations were performed considering wind velocities up to 50 m/s in order to evaluate the structural safety margin under extreme loading scenarios, significantly more severe than those expected during normal operation.

The obtained numerical results confirmed the adequacy of the proposed structural design as under the most critical loading condition, the maximum displacement of the monitoring station, shown in Fig 15(a), was equal to approximately 0.17 mm, occurring on the support of the solar panel, however the maximum displacement over the pole is around 0.15 mm in the middle region of the supporting structure, compatibility with the introduction of the gui-wire system fixed in the top position of the pole changing the constraint configuration, moving the maximal deformation from the upper extremity of the structure and reducing the amount of maximum excursion in presence of the imaging device, allowing deformation to remain sufficiently limited avoiding any significant variation of the camera orientation or degradation of the image acquisition process. Similarly, the maximum Von Mises stress, shown in Fig 15(b), reached 9,05 MPa, as expected, at the junction point between the pole and the middle of the support for the solar panel characterized by the maximal area exposed to the wind, remaining well below the yield strength of the adopted structural steel and resulting in a minimum safety factor, defined as the ratio between the yield strength and the maximum obtained stress, equal to 35,58 under static conditions. These results demonstrate that the supporting structure satisfies the required mechanical performance while preserving adequate safety margins under both operational and extreme environmental conditions.

Beyond static resistance, the long-term reliability of the monitoring station also depends on its dynamic behavior, for this reason a modal analysis was performed to identify the natural vibration modes of the complete structure and evaluate the possibility of resonance phenomena generated by environmental

excitations such as wind loading. The first natural frequency of the assembled monitoring station was found to be 13.85 Hz being a global combination of torsion and flexion mode as shown in Fig 15(c).

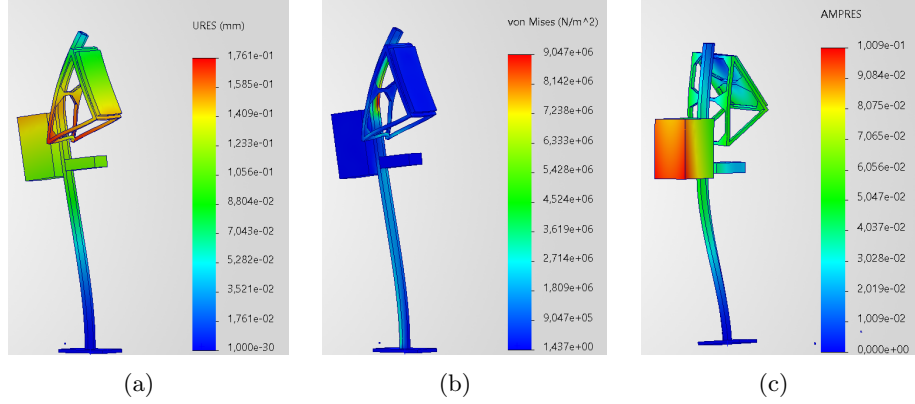


Figure 15: (a) Maximum displacement obtained from static structural analysis under combined gravity and wind loading. (b) Equivalent von Mises stress distribution obtained from static structural analysis. (c) First structural vibration modes obtained from modal analysis.

Successive modes are identified at 16.83 Hz, 28.54 Hz and 69.72 Hz being all global behavioral modes, while the subsequent vibration modes occurred at progressively higher frequencies characterized by local stimulated regions. Using the characteristic dimensions adopted for the exposed structural elements, the real historically recorded wind velocities up to approximately 18 m/s may generate vortex-shedding excitation frequencies within a range extending to approximately 36 Hz. Under this simplified excitation model, the first three global modes therefore fall within the potentially excitable frequency range. Considering instead the conservative marginal loads of 50 m/s, frequencies of up to around 100 Hz can be excited including also the fourth mode among the excitable ones. Although several natural frequencies fall within the theoretically excitable range, the harmonic response analysis, shown in Fig 16, indicates that the resulting stress amplification remains acceptable. The largest response amplification was obtained in correspondence with the first natural frequency, where the harmonic analysis predicted a peak equal to approximately 12.28 times the corresponding static response. The wind velocity associated with this excitation condition produced a static stress of approximately 6.23 MPa; applying the maximum amplification factor therefore resulted in an estimated peak stress of approximately 76.5 MPa. This value remains well below the adopted steel yield strength of approximately 322 MPa, corresponding to a safety factor of approximately 4.2. The remaining resonance peaks produced lower stress levels. Therefore, within the assumptions and excitation model adopted for the analysis, the dynamic response is not expected to compromise the structural integrity of the monitoring station or significantly affect the stability of the imaging system.

The influence of the guy wires on the global structural behavior was also quantitatively evaluated through comparative numerical analyses. Rather than being introduced solely as additional supporting elements, the guy wires were assessed in terms of their contribution to the overall stiffness of the monitoring station, indeed, simulations demonstrated a significant reduction of lateral displacements, being approximately halved and displaced in position to a less critical portion of the structure being the middle part, together with an increase in the overall structural rigidity, particularly under high wind loading conditions. Their

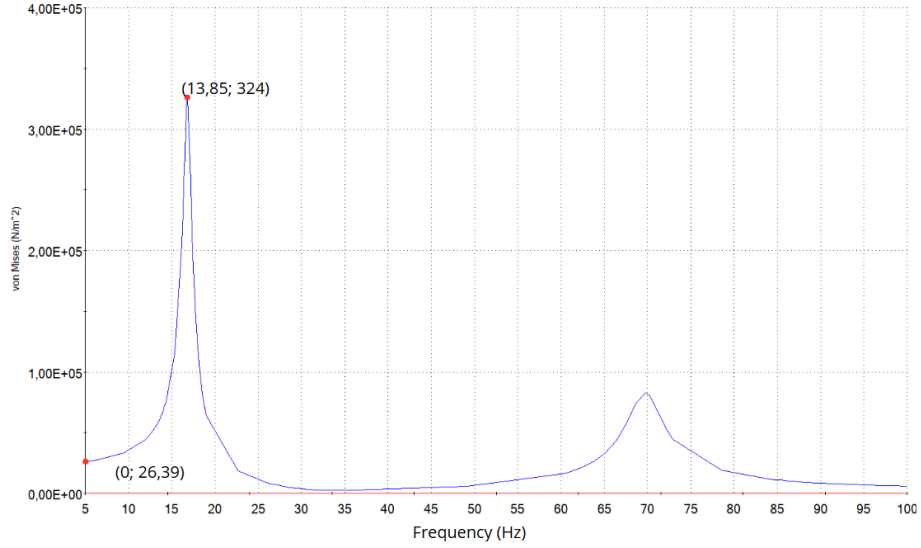


Figure 16: Frequency response of the supporting structure compared with the expected excitation frequencies.

contribution therefore proved essential in limiting structural deformations while simultaneously improving the dynamic behavior of the monitoring station without requiring substantial modifications to the primary supporting structure. Further analysis were performed to assess whether the initial estimated pre-load introduced artifacts in the simulations, however variations of the pre-load of one order of magnitude around 1000 N showed negligible effects over variations of the natural frequencies.

Overall, the numerical analyses confirmed that the adopted structural configuration satisfies the mechanical requirements identified for the monitoring campaign as the combined results of the static, modal and frequency response analyses demonstrate that the supporting structure provides adequate strength, stiffness and dynamic stability while also maintaining sufficient safety margins under conservative loading conditions. These characteristics ensure that the mechanical subsystem is capable of preserving the operational performance of the imaging, energy and communication subsystems throughout long-term autonomous deployment in the expected environmental conditions. Furthermore the structural displacement has been a critical factor not only with respect to the mechanical limits of the support but also in relation to the stability of the camera position, since excessive displacement could modify the acquisition geometry and consequently affect the quality and consistency of the image-analysis pipeline input.

5.5.3 Deployment and mechanical integration

The proposed mechanical architecture was ultimately translated into a fully assembled monitoring station through the integration of all the mechanical, electrical and imaging subsystems previously described. As previously illustrated, the assembly process was conceived to minimize installation complexity while ensuring that each subsystem could be mounted, inspected and, if necessary, replaced independently without affecting the remaining components and thanks to this modular approach both the initial deployment and future maintenance activities are simplified, particularly considering the limited accessibility of the

monitoring site. The installation sequence was organized according to the structural hierarchy of the monitoring station. Following the installation of the supporting pole provided by the local project partners, the solar panel preassembled in the laboratory to its support structure was first mounted using three stainless-steel mounting collars in order to freely orient the generator towards south, maximizing solar exposure. The electrical box was subsequently installed to fix the distance that the external cables should run from the solar panel to the box interior where they were connected with the rest of the electrical components placed in the box beforehand. After these two first components in order to avoid sensible oscillation of the pole topmost part the guy-wire system was installed connecting the top of the pole with three equally distanced points in the ground. Finally, the imaging device was mounted through the adjustable ball-head support and accurately aligned with the monitored river section before completing the electrical connections between it and the output of the DC-DC converter. The final structure can be observed in Fig 17. This assembly sequence minimizes installation time while avoiding dangerous excursions of the pole and allowing each subsystem to be tested independently before commissioning the complete monitoring station. Particular attention was dedicated to facilitating future maintenance operations without requiring complete disassembly of the installation as the electrical enclosure provides direct access to all power-management components, allowing inspection of the battery, charge controller, protection devices and electrical connections through the removable front panel. Similarly, the camera and photovoltaic module can be removed or replaced independently by acting exclusively on their dedicated mounting systems, reducing the duration of maintenance interventions whenever access to the monitoring site becomes possible. The use of commercially available fastening systems and standard mechanical components further simplifies replacement operations while improving the overall maintainability of the monitoring station.

The modular organization adopted throughout the mechanical design also facilitates the replication of the proposed monitoring solution in different deployment scenarios since the supporting structure, imaging subsystem, photovoltaic module and electrical enclosure constitute independent functional units, making future adaptations to different monitoring locations achievable primarily by modifying the supporting interfaces while preserving the overall system architecture ultimately reducing redesign effort and improving the scalability of the proposed solution for future monitoring campaigns.

The installation confirmed the practical feasibility of the proposed mechanical design, demonstrating that the structural architecture, electrical integration and subsystem arrangement developed throughout the engineering process can be effectively translated into a fully operational monitoring station representing the transition from the engineering design phase to the operational implementation of the proposed monitoring solution.

5.6 Image Analysis via Machine Learning

5.6.1 From acquired images to turbidity information

The subsystems presented in the previous sections guarantee the continuous acquisition and transmission of high-resolution riverbed images, but the raw images do not by themselves constitute the information the end users need. To deliver an actionable early-warning indicator of debris-flow susceptibility, the images must be automatically translated into a quantitative estimate of water turbidity, an optical proxy related to suspended sediment concentration and one of the earliest visually observable precursors of an incoming debris flow. This translation is the task of the image analysis subsystem, closing the loop between the acquisition hardware of Sections 5.3–5.5 and the decision-support information delivered to the end users.



Figure 17: Photo taken after installation completion on site of the complete structure.

The design faced two conflicting constraints: robustness to the wide variability of outdoor riverine conditions (illumination, viewpoint, weather), which a classical, hand-engineered pipeline struggles to generalize to, versus the monitoring station having only recently been installed, with no labeled field dataset yet available since turbidity labels require concurrent in-situ water sampling not carried out to date. Rather than postponing validation, a controlled laboratory smoke test was designed to assess, under fully controlled conditions, whether the proposed pipeline can in principle recover turbidity from photographs of the kind the deployed camera will produce.

5.6.2 Laboratory smoke test setting

The laboratory setup reproduces, at small and controllable scale, the essential characteristics of the field problem: a fixed camera photographing a body of water whose turbidity varies from clear to visually opaque. A transparent tank was filled with clean water and photographed from multiple camera positions and distances, a suspended material was then progressively added in 10 percentage-point increments from 0% to 100% (eleven levels per material), with the tank re-photographed after each addition. Three materials commonly responsible for river turbidity were used: bentonite clay, fine soil ("Dirt") and a white mineral powder. Images were additionally acquired from the front, left and right of the tank for every combination, introducing viewpoint variability that could mirror wind-induced vibration, repositioning or panoramic stitching in the field. The resulting dataset comprises 768 images spanning three materials,

eleven concentration levels, three camera positions and distances from 50 cm to 450 cm (Fig. 18).

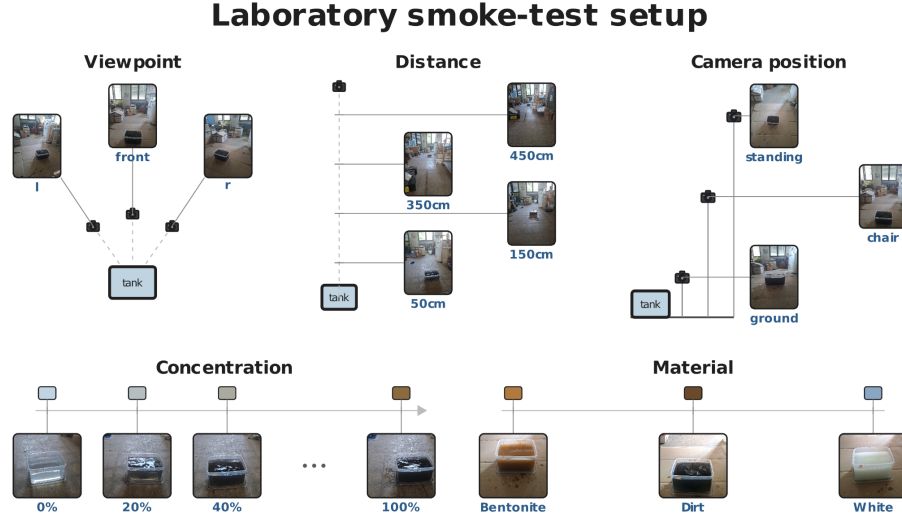


Figure 18: Overview of the laboratory smoke-test setup, illustrated with schematic diagrams paired with real photographs isolating each acquisition variable in turn while holding the other three fixed: viewpoint (l / front / r), distance (50 to 450 cm), camera position (ground / chair / standing) and material concentration (0-100% Dirt in 10% increments, 4 of 11 levels shown), the last of which already shows the visual saturation between mid- and high-range turbidity discussed in Section 5.6.4.

This laboratory setting was preferred over an early field trial for three reasons: the turbidity level is a precisely known and controlled quantity rather than an estimated one, the full range of turbidity including the rare but safety-critical near-opaque configurations can be sampled systematically and lastly it decouples pipeline development from the timeline of field data collection, letting both proceed in parallel. The approach naturally introduces a domain gap with the real deployment, discussed in Section 5.6.5.

5.6.3 Proposed image analysis pipeline

The core engineering choice was to avoid designing a dedicated feature extractor from scratch and instead reuse a pretrained foundation model to convert each photograph into a compact numerical representation, from which a lightweight regressor estimates turbidity. A foundation model denotes a large network trained once on a broad, task-agnostic data collection so that its representations can be reused with little or no further training across unrelated tasks [6]. This contrasts with training from randomly initialized weights, which typically needs tens of thousands of labeled examples, far more than the laboratory dataset provides. Reusing a representation learned on a vastly larger dataset than could ever be collected for this project makes it possible to obtain informative features even from 768 images, since only a small model must then be trained on top of them.

The foundation model adopted is DINOv2-giant [30], producing a 1536-dimensional representation per image. DINO ("self-distillation with no labels") [7] is a self-supervised method for vision transformers: the

network is shown two augmented views of the same image and trained so a "student" copy's representation of one view matches that of a slowly updated "teacher" copy for the other, forcing it to capture semantic content without any human-provided label.

Vision transformers divide an image into a grid of patches and apply self-attention [37] across the resulting sequence, letting every patch relate to every other regardless of distance [10]. A summary "CLS" token aggregates the whole image into a single vector; this 1536-dimensional CLS embedding is the representation used here. DINOv2 was chosen for its strong transfer performance on fine-grained texture and colour discrimination, and because its self-supervised objective, unlike language-supervised models, is not biased toward features describable in words. The backbone is kept frozen, making feature extraction a fixed step taking roughly 2.5 minutes for the full dataset on a single consumer GPU. Given the fixed embedding, the regressor predicts a three-element target vector, one entry per material, exactly one non-zero (all-zero for clean water). Framing this as a joint regression, rather than three independent regressors or a classify-then-regress pipeline, lets a single compact model estimate both severity and, via the non-zero entry's position, material type. A small multilayer perceptron with decreasing layer widths predicts this vector, deliberately kept shallow given the modest dataset size; a deeper model would risk memorizing the training examples. The network was trained with AdamW [23] under a cosine-annealed schedule [24] with early stopping. Because clean-water instances are comparatively rare, the training loss was weighted by inverse class frequency so their correct identification, arguably the most operationally important prediction, is not drowned out. Material is read off directly as the arg-max entry, with a small threshold below which the image is classified as clean water, avoiding a separate classifier (Fig. 19).

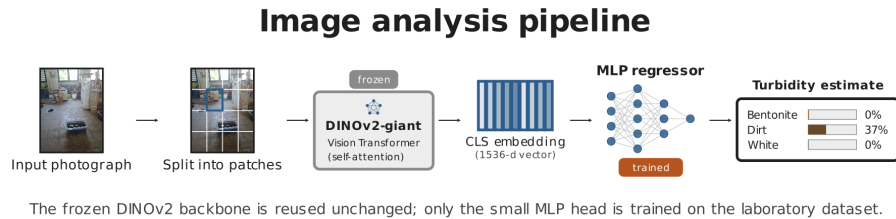


Figure 19: Overview of the proposed image analysis pipeline: an input photograph is processed by the frozen DINOv2-giant vision transformer into a 1536-dimensional CLS embedding, from which the trained MLP regressor estimates the percentage concentration of each candidate material.

The pipeline was evaluated with two complementary cross-validation strategies. Under the grouped split, entire turbidity configurations (same material and concentration, any camera position or viewpoint) are assigned exclusively to training or validation, measuring generalization to unseen turbidity levels, the more demanding and representative test of the field scenario. Under the random split, images are shuffled without regard to configuration so the same turbidity level can appear via different viewpoints in both sets measuring generalization across viewpoints of an already-seen level. Reporting both matters because, as shown below, a pipeline that performs well under one is not guaranteed to do so under the other.

5.6.4 Results

The tuned baseline, on the full 768-image dataset, achieved MSE 177.7 / MAE 6.59 percentage points / material accuracy 92.6% under the grouped split, and MSE 145.1 / MAE 5.89 / accuracy 93.2% under the

random split (Fig. 20).

An MAE below 7 points on a 0-100% target indicates that the frozen DINOv2 representation, despite never having seen turbidity data, retains enough visually relevant information for a simple downstream model to recover a fairly accurate estimate directly from a photograph.

The consistently higher grouped-split error, observed throughout every experiment below, confirms that generalizing to an unseen configuration is harder than generalizing across viewpoints of a familiar one, making the grouped split the more representative indicator of field performance.

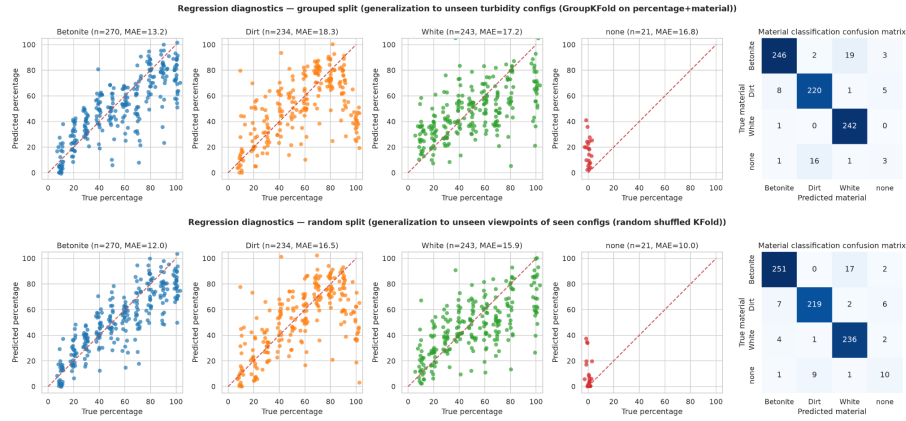


Figure 20: Baseline true-versus-predicted turbidity scatter plots (one panel per material, plus the clean-water "none" class) and material classification confusion matrix, under the grouped cross-validation strategy (top) and the random cross-validation strategy (bottom).

Breaking the error down by camera position, distance and viewpoint modifier showed viewing distance to be the dominant confound: MAE rises monotonically with distance, reaching ~31.5 points at standing/450cm versus 11-14 points at short range (Fig. 22).

This matches the intuition that turbidity cues (contrast loss, murky coloration) become harder to resolve as the imaged region shrinks within the frame. Excluding the two farthest configurations (350/450 cm) reduced error by 6-13% across both splits, confirming distance as an addressable confound (Fig. 23). The full dataset was nonetheless retained for all later experiments, since excluding hard viewing conditions would misrepresent the variability the deployed camera will face; this experiment is reported as diagnostic only.

An image augmentation strategy was then introduced to improve robustness without discarding data. Embedding-space augmentation (Gaussian noise, mixup [42]) did not help, suggesting the regressor was not meaningfully overfitting; augmentation was instead applied in pixel space, before DINOv2, via horizontal flip, color jitter and small rotation (Fig. 24).

Augmented views were embedded and appended to each fold's training set only, with validation always on original embeddings. This produced a clear, consistent improvement: MSE and MAE fell by roughly 13% and 11% across both splits (Fig. 25).

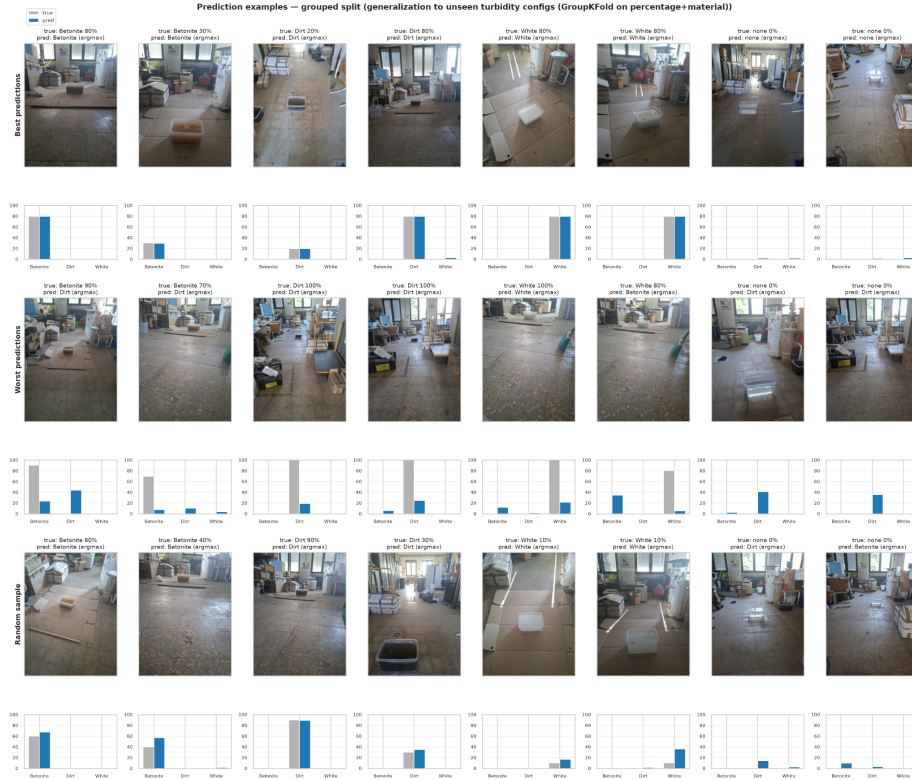


Figure 21: Example predictions under the grouped split: best predictions per material (top), the worst predictions per material (middle), and a random sample (bottom), each with the true (gray) and predicted (blue) concentration per material. The worst predictions cluster at high true percentages, an early sign of the saturation behavior discussed below.

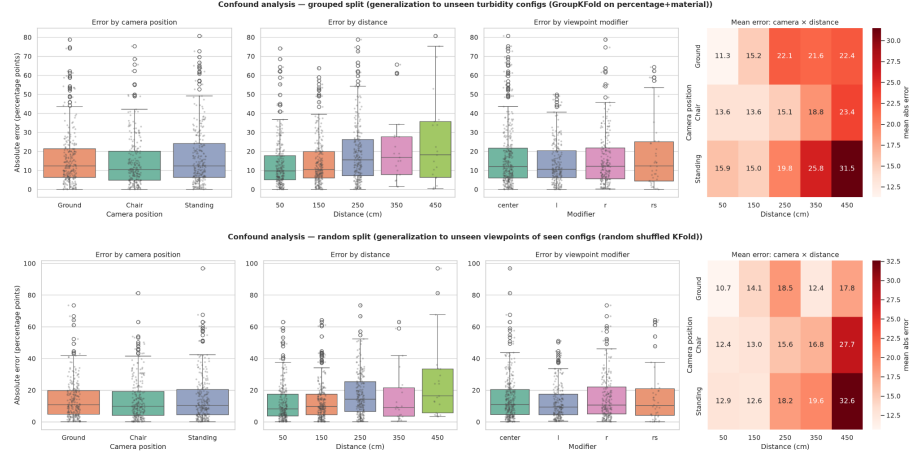


Figure 22: Regression error broken down by camera position, viewing distance and viewpoint modifier, under the grouped (top) and random (bottom) cross-validation strategies. Viewing distance is the dominant confound, with error rising monotonically from short range to 450 cm.



Figure 23: Regression MSE, MAE and material classification accuracy on the full dataset versus excluding the 350 cm and 450 cm configurations flagged by the confound analysis, for both cross-validation strategies.



Figure 24: Example original/augmented image pairs, showing the flip, colour jitter and small rotation transform applied in pixel space before DINOv2 feature extraction.

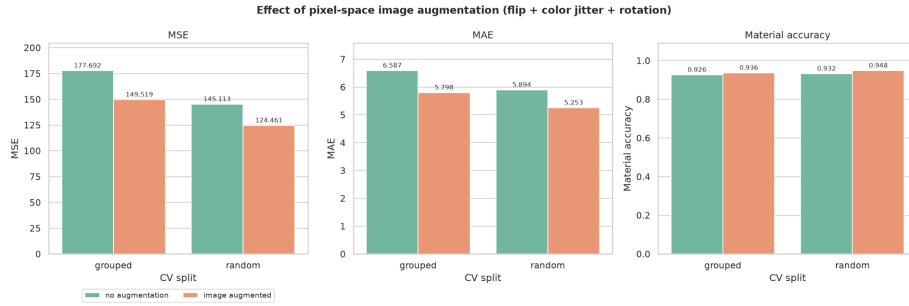


Figure 25: Regression MSE, MAE and material classification accuracy with versus without pixel-space image augmentation, for both cross-validation strategies.

A subsequent grid search over augmentation strength, evaluated jointly on both splits after an initial single-split search had picked a random-resized-crop variant that turned out to regress on the grouped split, identified stronger jitter with a larger rotation angle and no cropping as best (cropping was found to occasionally remove the tank from frame at close range) (Fig. 26). Overall, augmentation improved grouped MSE from 177.7 to 149.5 (-16%), MAE from 6.59 to 5.80 (-12%) and accuracy from 92.6% to 93.6%, with comparable ~14%/11% gains and an accuracy rise to 94.8% under the random split.

Despite these gains, every diagnostic plot, including the final scatter plots in Fig. 27, shows the same failure mode: systematic underestimation in the 90-100% range, where the tank becomes visually opaque and loses the transparency and coloration cues the model relies on. Neither distance filtering nor augmentation corrected this, suggesting a fundamental limitation of single-photo RGB turbidity estimation near saturation rather than a data or capacity issue. Addressing it likely requires a non-linear target transform or complementary features targeted at the near-opaque regime.

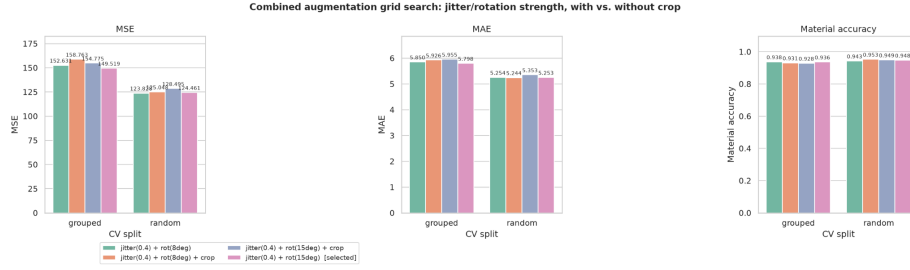


Figure 26: Combined augmentation grid search: the winning jitter/rotation transform stacked with a stronger rotation angle and/or random-resized cropping, evaluated on both cross-validation strategies. Cropping regresses in every combination tried, jitter(0.4) with 15-degree rotation and no crop wins overall and was adopted as the final configuration.

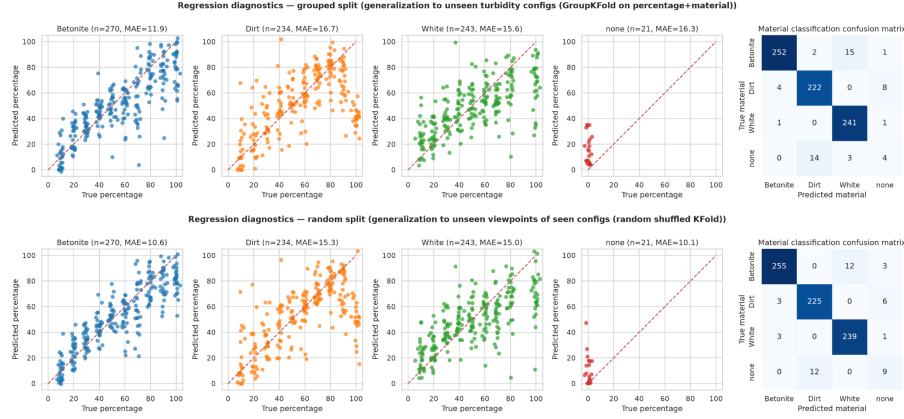


Figure 27: True-versus-predicted turbidity scatter plots and material classification confusion matrix for the final configuration, under the grouped (top) and random (bottom) cross-validation strategies. The systematic underestimation in the 90-100% range appears in every material panel as a flattening of the point cloud below the diagonal.

5.6.5 Relevance of the laboratory results to the field deployment

This experiment was never intended to produce a model directly deployable on the monitoring station; its purpose was to smoke-test the pipeline ahead of field data and surface its main limitations early. Several findings carry direct implications. An accurate regressor trained from only 768 images suggests a similarly compact model could be trained on comparably small field data, without the tens of thousands of examples training from scratch would require. Viewing distance being the dominant error source is directly relevant to the camera configuration of Section 5.3, chosen to hold a fixed ~ 7.7 mm ground resolution across the riverbed specifically to mitigate such degradation, so the imaging subsystem already partially compensates for a failure mode identified independently here.

The clear benefit of augmentation suggests an analogous, field-tailored strategy will remain valuable once field images are available, particularly while field data remains scarce. Most significantly, saturation at high turbidity matters operationally as reliably detecting the most severe events, precisely those most indicative of an incoming debris flow, is the system's primary purpose, so the early-warning logic should combine the regressed estimate with complementary indicators or a conservative threshold rather than the raw output alone.

The domain gap with the real site must still be acknowledged as a transparent, artificially lit tank at up to 450 cm differs from a natural, non-transparent riverbed photographed from a fixed elevated position at greater distances, and natural turbidity is compounded by flow velocity, foam and riverbed color. The laboratory results should therefore be read as evidence that the approach is methodologically sound and as a source of transferable design lessons, such as the value of augmentation and the criticality of the high-turbidity regime, rather than as a direct estimate of field accuracy. Validating the pipeline on genuine field images paired with in-situ turbidity measurements remains the necessary next step.

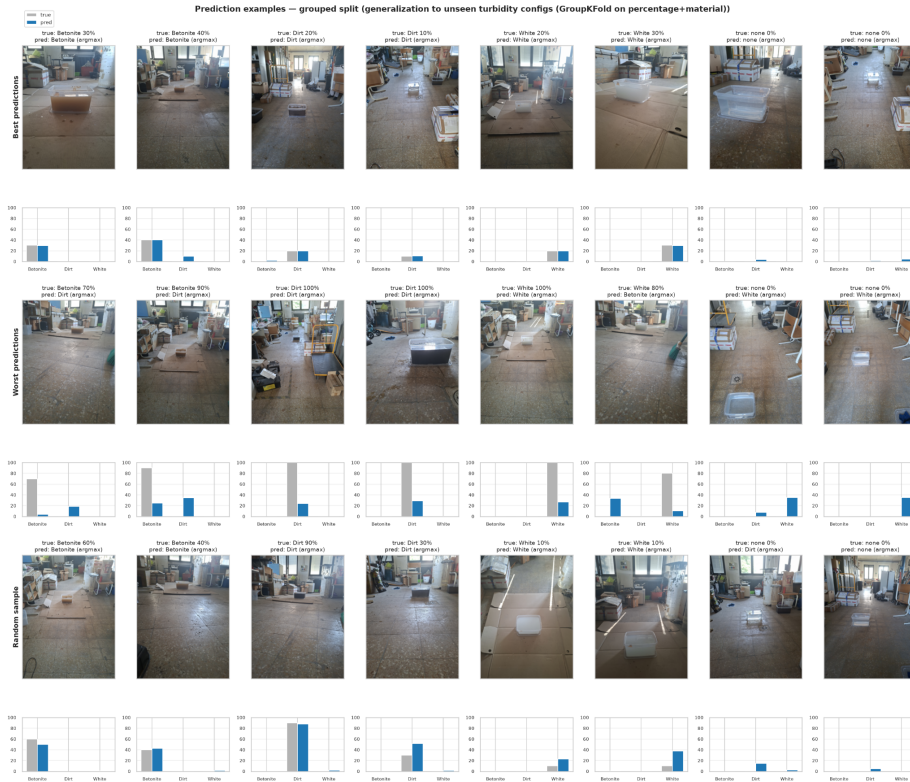


Figure 28: Example predictions for the final configuration under the grouped split, in the same best/worst/random format as Fig. 21. The worst predictions remain concentrated at high true percentages, consistent with the saturation behavior augmentation did not correct.

5.7 Final solution evaluation and considerations

5.7.1 System evaluation

Following the design and integration of the individual subsystems presented in the previous sections the proposed monitoring solution can now be evaluated at system level with respect to the specifications defined in Section 5.2. While the previous sections analyzed each subsystem independently of an engineering perspective, the objective of this section is to assess whether their integration results in a monitoring station capable of fulfilling the functional, operational and environmental objectives that originally motivated the project. From a functional perspective, the proposed solution satisfies all the principal capabilities identified during the system specification phase as it provides periodic acquisition of high-resolution observations over the entire monitored river section while the integrated LTE communication infrastructure guarantees the automatic transmission of every acquired image to the remote cloud platform without requiring manual data retrieval. The LTE communication workflow has been successfully verified under laboratory conditions, while long-term field validation remains part of the subsequent monitoring phase. The collected observations will therefore be immediately available for storage, remote inspection and subsequent processing through the Machine Learning workflow developed within the project, ensuring the continuity of the complete information pipeline from field acquisition to data analysis. The operational specifications are similarly addressed through the integration of the energetic, electrical and imaging subsystems. The autonomous photovoltaic power supply continuously supports the programmed acquisition schedule while eliminating the need for external electrical infrastructure. The selected acquisition strategy, combined with the automatic adaptation to sunrise and sunset conditions, enables continuous monitoring throughout the year while reducing unnecessary energy consumption while, at the same time, the integrated communication system and cloud-based management platform allow remote supervision of the monitoring campaign, significantly reducing maintenance activities and avoiding routine site visits solely for data collection. The proposed monitoring station also satisfies the environmental constraints associated with long-term deployment in a remote alpine environment as the mechanical support structure provides the required structural stability under the expected environmental loads while the protected electrical enclosure and the selected commercial components ensure compatibility with the harsh climatic conditions characterizing the installation site. Furthermore, the adopted energy management strategy explicitly accounts for the seasonal variability of the available solar resource, ensuring that the monitoring station remains operational even during the least favorable periods of the year.

An important outcome of the proposed design process is that the overall system performance cannot be attributed to the optimization of any individual subsystem alone. Instead, the final solution results from the coordinated integration of structural, energetic, electrical, imaging and computational elements, each contributing to the achievement of the overall monitoring objectives as several design choices required balancing competing requirements across different engineering disciplines, demonstrating that the effectiveness of the monitoring station depends primarily on the consistency of the complete system architecture rather than on the isolated performance of its individual components.

At the time of writing, the LTE communication architecture has been successfully tested under laboratory conditions but has not yet been validated during long-term field operation as during the initial deployment phase image acquisition and local system operation were verified on site while the intended LTE-based cloud transmission remains subject to subsequent field commissioning. Consequently, the communication subsystem should currently be considered laboratory validated rather than fully field validated.

Consequently also the whole processing pipeline has currently been tested only on laboratory acquired images even though extensive testing has been performed to verify the correct functioning of all the processes involved in the automatic retrieval and manipulation of the received images.

Table 7 summarizes the correspondence between the principal system specifications introduced in Section 5.2 and the engineering solutions adopted throughout the project highlighting that each functional, operational and environmental requirement identified during the preliminary design phase is directly addressed by one or more elements of the final monitoring station, confirming the consistency of the proposed solution with the original engineering objectives.

System specification	Evidence	Status
~8 m coverage	Camera FOV and installation geometry	Achieved
~1 cm resolution	Calculated ground resolution of 7.7 mm	Achieved
15 min acquisition	Camera operating configuration	Achieved
Autonomous power	PV and battery simulations	Verified by simulation
Structural stability	FEM/modal analyses	Numerically verified
Remote transmission	LTE architecture / deployment status	Laboratory tested
Turbidity estimation	Laboratory validation	Laboratory validated
Long-term operation	Design basis and numerical simulations	Not yet field-validated

Table 7: Summary of system requirements, supporting evidence, and validation status.

5.7.2 Deployment verification and system commissioning

The installation of the monitoring station provided the first opportunity to verify the practical implementation of the complete monitoring solution under real operating conditions and although the long-term monitoring campaign extends beyond the scope of the present work, the deployment activities allowed the correct integration of all subsystems to be assessed and confirmed that the proposed design can be effectively implemented outside the laboratory environment. Following the installation of the supporting structure, photovoltaic subsystem, electrical enclosure and imaging device, each subsystem was progressively commissioned according to the designed operating sequence. The electrical architecture was first energized and the correct operation of the photovoltaic charging system, battery management and DC/DC conversion stage was verified through the integrated monitoring interface of the charge controller and thanks to multimeter reading across different phases of the activation cycle. Once the power supply subsystem had been validated, the imaging device was connected and configured according to the operating parameters presented in Section 5.3 and after assessing its correct functioning it has been calibrated to observe the whole river section adjusting the ball head mechanism described in Section 5.3. Particular attention

was dedicated to verifying the interaction between the different subsystems rather than their individual operation. The monitoring station successfully demonstrated the capability to supply the camera through the designed electrical architecture, automatically execute the programmed acquisition schedule, establish the LTE communication link, verified in laboratory, and upload the acquired images to the cloud platform without requiring manual intervention (currently tested using Wi-Fi technology). These commissioning activities confirmed that the electrical, mechanical and communication interfaces developed throughout the design phase operate consistently when integrated into the complete monitoring station.

The deployment also provided the opportunity to verify several practical aspects that cannot be fully assessed through numerical analyses alone. The adopted mechanical layout facilitated cable routing, simplified access to the electrical components during installation and allowed the orientation of both the camera and the photovoltaic panel to be adjusted directly on site. Similarly, the internal arrangement of the electrical enclosure proved compatible with maintenance operations while preserving adequate separation between the different electrical components and cable paths. From an operational perspective, the successful installation demonstrated that the proposed monitoring solution can be deployed in a single day of work without requiring specialized infrastructure beyond the prepared support foundation. After being installed and configured, the monitoring station entered autonomous operation and successfully executed the defined image-acquisition schedule. As previously mentioned, image transmission and cloud-based data transfer were verified through the available communication configuration while the long-term LTE communication mode remains subject to subsequent field commissioning.

It should be emphasized that the commissioning activities performed at this stage constitute an operational verification of the implemented system rather than a complete performance validation as the objective was to demonstrate the correct integration and functionality of the monitoring station immediately after deployment, whereas the evaluation of long-term reliability, seasonal energy performance and monitoring effectiveness necessarily requires extended operation over the duration of the monitoring campaign. These aspects are therefore considered part of the future operational assessment rather than the initial deployment verification presented in this work.

6 Conclusion

6.1 Main Outcomes

The SENTINEL project resulted in the development and field deployment of an integrated autonomous monitoring system for the continuous optical observation of sediment dynamics in a glacier-fed alpine stream. The final architecture integrates a high-resolution imaging subsystem, autonomous photovoltaic energy generation and battery storage, a mechanically stabilized support structure, remote data communication capabilities, and a machine-learning-based image-analysis pipeline for turbidity estimation.

The structural subsystem was numerically verified under conservative static and dynamic loading conditions, confirming adequate stiffness and safety margins under the considered environmental actions. The energy subsystem was designed to sustain the expected operating profile throughout seasonal variations in solar availability, while field deployment confirmed the practical integration of the mechanical, electrical and imaging components and the feasibility of autonomous image acquisition under real installation conditions.

The image-analysis pipeline demonstrated the feasibility of estimating turbidity directly from optical observations under controlled laboratory conditions. Using DINOv2-based feature extraction and a lightweight regression model the final configuration achieved a grouped-validation mean absolute error of approximately 5.8 percentage points and a material-classification accuracy of approximately 93.6% confirming the technical potential of the proposed approach while providing a quantitative basis for subsequent field calibration demonstrating that the main engineering objectives of the project were addressed at different levels of validation. Autonomous image acquisition and the physical integration of the monitoring station were demonstrated during deployment, the structural and energetic requirements were assessed through numerical verification and the conversion of optical observations into quantitative turbidity estimates was demonstrated under controlled conditions. However, the long-term autonomous operation of the complete system and the quantitative accuracy of the monitoring output under real alpine conditions remain to be experimentally validated.

6.2 Limitations and Validation Requirements

Despite the successful integration and deployment of the monitoring station, the current system should be considered an operationally verified prototype rather than a fully validated sediment-monitoring or early-warning platform. The main limitation concerns the image-analysis subsystem, which has so far been validated only using controlled laboratory data. Consequently, the reported prediction accuracy cannot yet be assumed to represent performance under real alpine operating conditions.

Field operation introduces additional sources of variability, including illumination changes, shadows, reflections, precipitation, snow cover, atmospheric visibility, water velocity, foam, riverbed appearance and seasonal variations. Furthermore, the laboratory experiments revealed a systematic underestimation at very high turbidity levels, indicating a saturation effect in the near-opaque regime and suggesting that RGB image information alone may become insufficient to distinguish between extreme turbidity conditions.

Long-term reliability also remains to be demonstrated experimentally. Although structural and energetic performance were verified through numerical analyses and initial deployment testing, extended operation is required to assess seasonal photovoltaic performance, battery behavior under low-temperature conditions, long-term mechanical stability, component degradation and communication reliability. In addition, LTE

transmission has been verified in laboratory conditions, while full on-site implementation still requires completion of the SIM-based communication setup.

6.3 Future Development

Future development of SENTINEL should focus on transitioning the current prototype from an operationally verified monitoring station to a field-calibrated and quantitatively validated environmental monitoring system. The highest priority is the construction of a representative field dataset by pairing the images acquired by the installed camera with independent in-situ measurements of turbidity and, where possible, suspended sediment concentration. These measurements should cover different hydrological conditions, seasonal periods, illumination scenarios and sediment loads in order to quantify the domain gap identified between the laboratory experiments and the real deployment environment. The resulting dataset should then be used to recalibrate and retrain the image-analysis pipeline, establish confidence intervals for the predicted turbidity values and determine the actual detection limits and uncertainty of the system under field conditions. Particular attention should be dedicated to the high-turbidity regime, where the laboratory experiments showed systematic underestimation due to optical saturation. Future iterations should investigate dedicated modeling strategies for this range, including non-linear target transformations, regime-specific models or the integration of complementary environmental indicators. The turbidity estimate should therefore evolve from a standalone prediction into one component of a broader monitoring framework that can incorporate variables such as precipitation, water level, temperature and other hydrological or geomorphological measurements. Such multimodal integration would improve the interpretation of anomalous sediment-transport events and reduce the dependence of the monitoring output on RGB information alone. In parallel, the complete station should undergo long-duration operational validation over multiple seasonal cycles. This phase should verify the real photovoltaic energy balance, battery performance during prolonged low-temperature periods, mechanical stability after repeated wind and weather exposure, communication reliability and the persistence of camera framing and image quality over time. The completion of the on-site LTE configuration is therefore an immediate technical requirement, after which the station can support continuous remote data transfer and systematic performance monitoring. During the first year after installation maintenance frequency, component degradation and system downtime should also be recorded to determine whether the low-maintenance and long-term autonomy objectives are achieved in practice.

Once sufficient field data and operational history are available, the system could be extended toward a decision-support and anomaly-detection framework. Rather than relying on a single turbidity threshold, future developments could identify statistically significant deviations from normal seasonal behavior and combine them with complementary environmental observations to recognize potentially critical sediment-mobilization events. At that stage, the applicability of SENTINEL to hazard assessment and, eventually, early-warning applications could be evaluated quantitatively. The modular architecture developed within the project also provides the basis for replication at additional glacier-fed streams, allowing the robustness, transferability and scalability of the proposed monitoring approach to be assessed across different alpine environments.

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A Appendix

A.1 Project Development and Timeline

The development of the project followed the overall structure and milestones defined for the ASP activity, while the scope and organization of the work evolved during the project in response to changes in the project objectives and the progressive definition of the requirements. After the team organization and first stages of the project a major change in the scope occurred in November of the first year, when the academic tutors had to redefine the application to be addressed, after the municipality rejected the request of installation of the system the project was originally intended to develop, resulting in a significant reorientation of the activities performed by the team and requiring the previously initiated work to be disregarded to focus towards the development of the final monitoring solution. A further relevant milestone occurred in mid-May, when the final set of system requirements became available, allowing the design activities to be progressively translated into a concrete system architecture, followed by the assembly and commissioning activities. The Gantt chart, presented on page 68 therefore reflects a relatively concentrated development and integration phase during the final part of the project.

These changes are reported here as elements of the project context rather than as deviations from the intended workflow as they affected the sequence and distribution of the activities performed by the team and consequently form part of the organizational framework within which the final solution was developed.

A.2 Team Organization and Work Allocation

At the beginning of the project, the team adopted the division of responsibilities described in Chapter 2 but such organization was introduced to provide a clear allocation of responsibilities and to allow the different aspects of the multidisciplinary project to be addressed in parallel.

The organization was nevertheless deliberately maintained flexible throughout the project. Rather than operating as two strictly separated groups, team members progressively adjusted their focus according to the workload associated with the different phases of development allowing additional effort to be directed towards the activities requiring greater attention at a given stage while maintaining a shared understanding of the work performed by the different parts of the team. This approach was particularly relevant for a multidisciplinary project in which the mechanical, electrical, imaging and computational aspects were strongly interconnected as maintaining knowledge across the different activities reduced the risk of creating isolated work packages whose interfaces could become difficult to manage during integration while also allowing team members to support activities outside their initial area of responsibility when required. The flexible allocation of work also made it possible to accommodate the different academic commitments of the team members during the project allowing the distribution to be adapted to the availability of individual members without compromising continuity in the development of the overall system.

A.3 Geographical and Logistical Organization

The physical location of the laboratory in Milan introduced an additional organizational constraint. During part of the project, some team members were located in Turin while others were participating in Erasmus activities abroad and consequently, the availability of team members for laboratory-based activities varied throughout the project. The work was therefore distributed not only according to technical responsibilities but also according to the type of activity and the physical presence required allowing team members who

were not regularly present in Milan to contribute to the project through activities that could be performed remotely, with a particular focus on organizational, administrative and coordination tasks while those members who were physically present in Milan concentrated to a greater extent on laboratory activities requiring direct access to the equipment and workspace.

This distinction became particularly relevant during the final assembly and integration of the system when the members present in Milan performed most of the manual activities required to assemble the structure, integrate the electrical components and install the imaging system, while the other members continued to contribute through activities that did not require physical access to the laboratory. The resulting organization allowed the team to maintain a continuous contribution from all members despite the geographical distribution of the group.

A.4 Budget and Resource Management

The project expenditure can be broadly divided into two main categories, namely field activities and resources required for the construction and integration of the monitoring system. The first category consisted of the two field trips performed during the project. The first expedition took place at the beginning of the project and was primarily aimed at understanding the characteristics of the deployment environment, assessing the practical constraints of the site and determining the requirements that the final solution would have to address. The second expedition was instead conducted during the final phase of the project and was dedicated to the installation of the developed system and its initial commissioning on the field, performed in July 2026. The second category comprised the materials and equipment required to construct and integrate the solution. Whenever possible, existing resources available in the laboratory or provided by the professors or team members were used while new purchases were made only when the required tools were not otherwise available and their acquisition was necessary to complete the development activities. These expenditures included both raw materials and components required for the physical construction of the system as well as tools and equipment intended to remain available in the laboratory after completion of the project.

This approach allowed the available budget to be concentrated on resources directly contributing to the development, assembly and deployment while avoiding unnecessary duplication of equipment already available within the project environment.

SENTINEL PROJECT — Master Gantt Chart Timeline (May 2025 - September 2026)

